

Lamport Proof Review

Too Late to Coordinate:
When Local Information Beats Global Sharing

Review of the six-page letter by Scott Moeller and Bhaskar Krishnamachari

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Main conclusion

The seven formal results are supported by the submitted arguments under the stated model. The forward review is **PASS**. The conclusion-first review is **FOLLOWS**. No critical, major, or minor proof issue, circular argument, or unresolved validity obligation was identified.

Theorem 1 checks the team optimality equations directly under each agent’s full local observation history and the common delayed history. Its central covariance identity gives the required conditional means for every intermediate observation time. The report checks that identity, its unequal-variance counterpart, all endpoint and refresh calculations, and the discrete-time and multirate consequences.

The distinction between the two communication comparisons is important. Fresh local information can outperform a *delayed pool used alone*. A hybrid that retains fresh local observations has strictly positive pooling value at every finite age. The no-refresh threshold for the hybrid concerns its *communication-inclusive average cost*.

The accompanying package is prepared for <https://github.com/ANRGUSC/TINA>. The source identifiers below specify the exact manuscript covered by this report.

This report follows the `convert-lamport`, `forward-lamport`, and `reverse-lamport` skills from the WWresearch Lamport Proof project. It is a model-assisted mathematical audit. No Lean, Coq, Isabelle, or other proof kernel was used.

ID	Result	Forward audit	Reverse audit
L	Lemma 1: Team optimality	PASS	FOLLOWS
P	Proposition 1: Fresh endpoints	PASS	FOLLOWS
D	Corollary 1: The delay crossover	PASS	FOLLOWS
H	Theorem 1: Hybrid policy and exponential value	PASS	FOLLOWS
U	Proposition 2: Unequal sensor quality	PASS	FOLLOWS
C	Corollary 2: Additive coordination value	PASS	FOLLOWS
R	Theorem 2: Optimal refresh period	PASS	FOLLOWS

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Part I

Source-preserving Lamport conversion

1 Rendering status

SOURCE-MAPPED. All seven formal proofs have an auditable rendering. The source inventory partitions their proof text into 67 consecutive segments. The hierarchy contains 73 labeled steps, including case boundaries and closing steps. Each source segment is placed, and each rendered step has a source entry.

This status concerns source mapping only; it is not a validity verdict.

The conversion retains the paper’s route. The hybrid proof uses information enlargement, Gaussian projection on the full history, and the necessary and sufficient team optimality equations. The refresh proof uses an interval-by-interval lower bound. The detailed calculations in Part II check the operations supplied by the source while keeping their provenance separate from the source rendering.

2 Frozen source contract

Here, “frozen” means that the document and assumptions being reviewed are fixed. It prevents a proof from being judged against a silently revised theorem. The target is the current six-page letter in `source/letter.pdf` and its matching `source/letter.tex`. The broader arXiv paper is not used to supply missing proof steps. The manuscript itself has not been edited by this review.

2.1 Model and conventions

There are $n \geq 2$ agents. Agent i observes $Y_i(t) = X(t) + E_i(t)$. The signal X and the sensor disturbances E_i are independent mean-zero stationary Gaussian processes. In the scalar equal-quality model,

$$\text{Cov}(X(t), X(r)) = ae^{-\lambda|t-r|}, \quad \text{Cov}(E_i(t), E_i(r)) = be^{-\lambda|t-r|},$$

with $a, b, \lambda > 0$. The rate is common to the signal and every disturbance. The expected team loss is

$$J(u) = \mathbb{E} \left[\frac{1}{n} \sum_i (u_i - X)^2 + \frac{\kappa}{n} \sum_i (u_i - \bar{u})^2 \right], \quad \bar{u} = \frac{1}{n} \sum_i u_i, \quad \kappa \geq 0. \quad (\text{A1})$$

Actions do not affect the signal or observations. Each action must be square-integrable and measurable with respect to its assigned information. All optimality claims therefore include nonlinear and history-dependent policies.

The local history is $\mathcal{F}_i(t) = \sigma\{Y_i(r) : r \leq t\}$ and the complete history is $\mathcal{F}(t) = \bigvee_i \mathcal{F}_i(t)$. Pooling time $s = t - \tau$ is in the past, so $\tau \geq 0$. The hybrid information is

$$\mathcal{H}_i(t; s) = \mathcal{F}_i(t) \vee \sigma\{\bar{Y}(r) : r \leq s\}.$$

These are sigma-fields generated by the indicated observations. Independence of a Gaussian residual from a whole history is checked against every finite subfamily of that history. The generated sigma-field statement follows by the usual extension from cylinder events. It does not assume that a finite numerical grid equals the full history.

The following conventions make the source’s notation explicit without adding model restrictions. Agent and component counts are finite positive integers. Ages and latencies are finite nonnegative real numbers. A “nonzero” policy perturbation is nonzero as an L^2 equivalence class. The long-run cost is the limiting upper average of *expected* accumulated loss and communication cost. The refresh proof starts without a pooled update, as stated by its zero-gain initial interval. A fixed initial summary would change only a bounded transient, not the long-run optimum.

2.2 Accepted mathematical background and source assumptions

The audit uses the elementary rules and named Gaussian, conditional-expectation, and convexity facts that the source invokes. Their precise forms are listed below so that each application has an explicit domain and conclusion. The source identifies Lemma 1 as a specialization of Radner’s stationarity condition and supplies its own variational proof, which is checked here. The related-work citations do not supply missing steps in any audited proof.

ID	Exact background rule or assumption
A-G	$n \geq 2$, $a, b, \lambda > 0$, $\kappa \geq 0$; mutually independent, stationary, mean-zero Gaussian signal and disturbances with the common exponential covariance (1).
A-L2	Admissible actions and perturbations are measurable with respect to their assigned information and have finite second moment. $X \in L^2$ under (1).
A-U	The A-G assumptions hold with disturbance variances $b_i > 0$ and unchanged common rate. The definitions C_i, θ are those in (23).
A-C	A finite number of mutually independent component processes with weights $w_m > 0$, componentwise common rates $\lambda_m > 0$, and ages $\tau_m \geq 0$.
A-R	Fixed finite latency, exact joint vectors of componentwise pooled means sampled at the same time, price $c > 0$, no queueing, and schedules independent of the sensor processes. Local finiteness means finitely many transmissions in each bounded interval. The proof uses an initially unshared state.
D-loss	$\ell(u, X) = n^{-1} \sum_i (u_i - X)^2 + \kappa n^{-1} \sum_i (u_i - \bar{u})^2$ and $\bar{u} = n^{-1} \sum_i u_i$.
D-Y	$Y_i = X + E_i$ and $\bar{Y} = n^{-1} \sum_i Y_i$.
D-vd	$v = a + b/n$ and $d = a + b[1 + \kappa(1 - 1/n)]$.
D-info	The local, pooled, delayed-only, and hybrid information fields are exactly those in Section II-B of the letter.
D-refresh	$\gamma_m = 2\lambda_m$, $A_m = w_m \Delta_m e^{-\gamma_m \delta}$, $B(T) = \sum_m A_m (1 - e^{-\gamma_m T})/\gamma_m$, and $C(T) = J_L^{\text{tot}} + [c - B(T)]/T$.
B-G	Gaussian regression and orthogonality: a jointly Gaussian residual with zero cross-covariance with every observation is independent of the observation sigma-field. For a nonsingular Gaussian observation vector, the conditional mean is its linear regression; the residual covariance is the corresponding Schur complement.
B-CE	Conditional expectation is the L^2 orthogonal projection. It minimizes mean-square error, is characterized by testing against measurable L^2 variables, satisfies the tower property, and is an L^2 contraction.
B-OU	For the stated stationary common-rate Gaussian process, the innovation over $[s, t]$ is independent of the past and has covariance $(1 - e^{-2\lambda(t-s)})\Sigma$. Its cross-covariances are computed directly from (1).
B-J	Conditional Jensen: $\phi(\mathbb{E}[U \mathcal{A}]) \leq \mathbb{E}[\phi(U) \mathcal{A}]$ for an integrable vector and convex function when the expectations exist. The audit checks information feasibility separately.

ID	Exact background rule or assumption
B-calc	Finite-dimensional real algebra and order, differentiation, integration of exponentials, and the intermediate value theorem for continuous scalar functions, with signs and domains checked at each use.
E-eta	The globally quantified residual statement accompanying source equation (11): $\eta(t) = X(t) - (a/v)\bar{Y}(t)$ is independent of the entire sensor process and has variance P_n . Theorem 1 explicitly cites this source result. Its covariance calculation is checked separately in K-P.

2.3 Scope of earlier results

The seven proofs are reviewed as separate theorem units. A later proof cites the public result L, P, or H, rather than a private descendant of a closed Lamport subproof. The globally quantified residual property accompanying source equation (11) is a special explicit source dependency, recorded as **E-eta**. Theorem 1 cites that equation, and the audit checks its covariance argument independently under the global model. No local case assumption or witness is exported with it.

2.4 Lemma 1: Team optimality

For the information fields $\mathcal{G}_i$, optimize over $u_i \in L^2(\mathcal{G}_i)$, with $X \in L^2$, $n \geq 2$, and $\kappa \geq 0$. A feasible u is optimal exactly when

$$(1 + \kappa)u_i - \kappa\mathbb{E}[\bar{u} \mid \mathcal{G}_i] = \mathbb{E}[X \mid \mathcal{G}_i] \quad (i = 1, \dots, n).$$

Two optimal feasible policies agree almost surely. Source: letter (3), (5), lines 102–112.

2.5 Proposition 1: Fresh endpoints

Under (1)–(3), let $v = a + b/n$ and $d = a + b[1 + \kappa(1 - 1/n)]$. For complete local and pooled histories,

$$P_1 = \frac{ab}{a+b}, \quad P_n = \frac{ab}{b+na}, \quad u_i^L = \frac{a}{d}Y_i(t), \quad u_i^P = \frac{a}{v}\bar{Y}(t).$$

The optimal losses are $J_L = a - a^2/d$ and $J_P = P_n = a - a^2/v$, and $\Delta = J_L - J_P = a^2(d - v)/(vd) > 0$. Source: letter (6)–(11), lines 117–145.

2.6 Corollary 1: The delay crossover

Let $\tau \geq 0$ and $s = t - \tau$. Every agent has $\mathcal{F}(s)$ and no post- s measurements. The optimal common action is $u_i^D = e^{-\lambda\tau}(a/v)\bar{Y}(s)$ and

$$J_D(\tau) = a - \frac{a^2}{v}e^{-2\lambda\tau}.$$

Then $J_L < J_D(\tau)$ exactly when $\tau > (2\lambda)^{-1} \log(d/v)$, with equality at that age. Source: letter (13), (14), lines 155–170.

2.7 Theorem 1: Hybrid policy and exponential value

Under (1)–(4), put $\rho = e^{-\lambda\tau}$ and $s = t - \tau$. The unique optimal hybrid action is

$$u_i^H = \rho \frac{a}{v} \bar{Y}(s) + \frac{a}{d} [Y_i(t) - \rho Y_i(s)].$$

It also attains the optimum with the larger information $\mathcal{F}(s) \vee \mathcal{F}_i(t)$. Its posterior variance and team loss satisfy

$$P_H = \rho^2 P_n + (1 - \rho^2) P_1, \quad J_H = \rho^2 J_P + (1 - \rho^2) J_L,$$

and $J_L - J_H = \Delta e^{-2\lambda\tau}$. Source: letter (15)–(21), lines 175–214.

2.8 Proposition 2: Unequal sensor quality

Replace b by $b_i > 0$, retaining independence, stationarity, Gaussianity, and a common temporal rate. Define

$$C_i = (1 + \kappa)a + b_i[1 + \kappa(1 - 1/n)], \quad \theta = n^{-1} \sum_i C_i^{-1}.$$

Then $k_i = a/[C_i(1 - \kappa a\theta)]$, $J_L = a - a^2\theta/(1 - \kappa a\theta)$,

$$P = (a^{-1} + \sum_i b_i^{-1})^{-1}, \quad \mu_P(t) = P \sum_i Y_i(t)/b_i, \quad J_P = P.$$

The hybrid action is $u_i^H = \rho\mu_P(s) + k_i[Y_i(t) - \rho Y_i(s)]$ and $J_H = \rho^2 J_P + (1 - \rho^2) J_L$. Source: letter (23)–(25), lines 230–248.

2.9 Corollary 2: Additive coordination value

There are finitely many independent components $m = 1, \dots, M$ and positive weights w_m . Within component m , the signal and all disturbances share $\lambda_m > 0$. Each component has a pooled summary of age $\tau_m \geq 0$. Then

$$J_L^{\text{tot}} - J_H^{\text{tot}} = \sum_m w_m \Delta_m e^{-2\lambda_m \tau_m}, \quad J_L^{\text{tot}} = \sum_m w_m J_{L,m},$$

and each component uses the corresponding hybrid policy (15). Source: letter (26), lines 252–261.

2.10 Theorem 2: Optimal refresh period

Assume fixed finite latency $\delta \geq 0$, an exact joint pooled-vector refresh with price $c > 0$, and locally finite schedules independent of all signal and disturbance processes. Let $\gamma_m = 2\lambda_m$, $A_m = w_m \Delta_m e^{-\gamma_m \delta} > 0$,

$$B(T) = \sum_m \frac{A_m}{\gamma_m} (1 - e^{-\gamma_m T}), \quad c_{\text{crit}} = \sum_m A_m / \gamma_m.$$

For $c \geq c_{\text{crit}}$, no refreshing minimizes the limiting upper average cost at J_L^{tot} . For $0 < c < c_{\text{crit}}$, a periodic schedule is optimal among all admissible schedules. Its unique period solves $B(T^*) - T^* B'(T^*) = c$, and its cost is $J_L^{\text{tot}} - \sum_m A_m e^{-\gamma_m T^*}$. Source: letter (28)–(33), lines 284–311.

3 Source inventory

The source IDs below cover every proof-bearing span of the seven formal proofs in source order. The line numbers refer to the unchanged `source/letter.tex`. The original text of every span is also stored in `frozen_conversion.json`. Several rendered steps may share one source span when a sentence supplies both a mathematical statement and its case or closing structure.

Source ID	Lines	Source claim or transition	Placement
SRC-001	111–111	For a feasible perturbation w , the objective difference has linear term $2n^{-1} \sum_i \mathbb{E}[w_i((1+\kappa)u_i - \kappa\bar{u} - X)]$ and a quadratic remainder.	L.1
SRC-002	111–111	The linear term vanishes for every feasible w if and only if the conditional normal equations (5) hold.	L.2
SRC-003	111–111	The remainder is $n^{-1} \sum_i \mathbb{E}[w_i^2] + \kappa n^{-1} \sum_i \mathbb{E}[(w_i - \bar{w})^2]$, positive whenever w is nonzero in L^2 .	L.3
SRC-004	111–112	Q.E.D. Necessity, sufficiency, and uniqueness follow.	L.4
SRC-005	134–134	$X(t) - aY_i(t)/(a+b)$ has zero covariance with every $Y_i(r)$ for $r \leq t$, so it is independent of the local history.	P.1
SRC-006	134–138	Define $\eta(t) = X(t) - (a/v)\bar{Y}(t)$. It is uncorrelated with every $Y_j(r)$ and independent of the entire sensor process.	P.2
SRC-007	138–140	The residual variances are $P_1 = ab/(a+b)$ and $P_n = ab/(b+na)$. The corresponding conditional means are $aY_i(t)/(a+b)$ and $(a/v)\bar{Y}(t)$.	P.3
SRC-008	140–142	For the candidate $u_i = kY_i(t)$, $\mathbb{E}[\bar{u} \mathcal{F}_i(t)] = kvY_i(t)/(a+b)$.	P.4
SRC-009	142–144	Equation (5) gives $k[(1+\kappa)(a+b) - \kappa v] = a$. The bracket equals d , so $k = a/d$.	P.5
SRC-010	144–144	The local loss is $a - 2ak + dk^2 = a - a^2/d$.	P.6
SRC-011	144–144	With pooled information all agents use the common conditional mean, with zero disagreement and loss P_n .	P.7
SRC-012	144–144	The displayed feasible policies are optimal by Lemma 1.	P.8
SRC-013	144–145	$d - v = b(1+\kappa)(1 - 1/n) > 0$ and subtraction yields the formula for Δ .	P.9, P.10
SRC-014	168–169	The Markov property and Proposition 1 give $\mathbb{E}[X(t) \mathcal{F}(s)] = e^{-\lambda\tau}(a/v)\bar{Y}(s)$.	D.1
SRC-015	169–169	Taking this common mean minimizes tracking loss and yields zero disagreement, hence minimizes the team loss.	D.2
SRC-016	169–169	The error variance is the displayed $J_D(\tau)$.	D.3
SRC-017	169–169	Comparison with J_L yields $J_L < J_D$ if and only if $e^{-2\lambda\tau} < v/d$.	D.4
SRC-018	169–170	Taking logarithms gives the exact threshold (14), including its equality case.	D.5, D.6
SRC-019	192–192	CASE $\tau = 0$. The formula is the pooled policy.	H.1
SRC-020	192–192	CASE $\tau > 0$. Prove the remaining claims under this assumption.	H.2
SRC-021	192–196	Enlarge each information field to $\mathcal{G}_i = \mathcal{F}(s) \vee \mathcal{F}_i(t)$ and define $Z_i = Y_i(t) - \rho Y_i(s)$, $h = a/v$, $k = a/d$.	H.2.1
SRC-022	196–196	Z is independent of $\mathcal{F}(s)$ and has covariance $(1 - \rho^2)(a\mathbf{1}\mathbf{1}^\top + bI)$.	H.2.2
SRC-023	196–202	$X(t) = \rho h\bar{Y}(s) + h\bar{Z} + \eta(t)$, where $\eta(t)$ is independent of all sensor data and has variance P_n .	H.2.3
SRC-024	202–206	For $s < r \leq t$, define $Z_i(r) = Y_i(r) - e^{-\lambda(r-s)}Y_i(s)$. Then $\text{Cov}(\bar{Z}, Z_i(r)) = [v/(a+b)] \text{Cov}(Z_i, Z_i(r))$.	H.2.4
SRC-025	206–206	Both covariances have the temporal factor $e^{-\lambda(t-r)}(1 - e^{-2\lambda(r-s)})$.	H.2.5
SRC-026	206–206	$\bar{Z} - vZ_i/(a+b)$ is orthogonal to the entire local innovation path and to $\mathcal{F}(s)$.	H.2.6
SRC-027	206–206	Gaussianity gives $\mathbb{E}[\bar{Z} \mathcal{G}_i] = vZ_i/(a+b)$.	H.2.7
SRC-028	206–211	Independence of $\eta(t)$ gives $\mathbb{E}[h\bar{Z} + \eta(t) \mathcal{G}_i] = aZ_i/(a+b)$ and $\mathbb{E}[k\bar{Z} \mathcal{G}_i] = kvZ_i/(a+b)$.	H.2.8

Source ID	Lines	Source claim or transition	Placement
SRC-029	211–211	Subtract the common prediction. Since $(1 + \kappa)(a + b) - \kappa v = d$, the correction kZ_i satisfies (5) under the full path information. Lemma 1 certifies optimality.	H.2.9
SRC-030	211–213	The policy (15) is feasible under the hybrid field (4), so it attains the same value as complete delayed sharing.	H.2.10
SRC-031	213–213	The irreducible variance is P_n and the remaining quadratic cost scales by $1 - \rho^2$. Thus $J_H = P_n + (1 - \rho^2)(J_L - P_n)$.	H.2.11
SRC-032	213–213	The enlarged-information posterior mean is $\rho h\bar{Y}(s) + aZ_i/(a + b)$. It is hybrid-feasible and has error variance $P_n + (1 - \rho^2)(P_1 - P_n)$.	H.2.12
SRC-033	213–214	Subtracting the cost from J_L yields $\Delta e^{-2\lambda\tau}$.	H.2.13, H.2.14, H.3
SRC-034	245–245	Substitution into (5) gives $C_i k_i - \kappa a \bar{k} = a$, where $\bar{k} = n^{-1} \sum_i k_i$.	U.1
SRC-035	245–245	Averaging $k_i = (a + \kappa a \bar{k})/C_i$ yields the displayed coefficients.	U.2
SRC-036	245–245	The denominator is positive because $C_i > (1 + \kappa)a$.	U.3
SRC-037	245–245	Lemma 1 certifies optimality of the local policy.	U.4
SRC-038	245–245	The quadratic term equals $a\bar{k}$, so $J_L = a - a\bar{k}$.	U.5
SRC-039	245–247	Gaussian conditioning gives P and μ_P .	U.6
SRC-040	247–247	$X(t) - \mu_P(t)$ is independent of the entire sensor process because all temporal covariances share one factor.	U.7
SRC-041	247–247	Innovations have covariance $(1 - \rho^2)\Sigma$, where $\Sigma = a\mathbf{1}\mathbf{1}^\top + \text{diag}(b_i)$.	U.8
SRC-042	247–247	For $\mathcal{G}_i = \mathcal{F}(s) \vee \mathcal{F}_i(t)$, the covariance argument in (21) gives $\mathbb{E}[Z_j \mathcal{G}_i] = \Sigma_{ji} Z_i / \Sigma_{ii}$.	U.9
SRC-043	247–247	After subtracting $\rho\mu_P(s)$, the target has conditional mean $aZ_i/(a + b_i)$.	U.10
SRC-044	247–247	The residual normal equations are the fresh-local equations, so $k_i Z_i$ is optimal.	U.11
SRC-045	247–247	Scaling the remaining quadratic cost by $1 - \rho^2$ gives $J_H = \rho^2 J_P + (1 - \rho^2) J_L$.	U.12
SRC-046	247–248	Q.E.D.	U.13
SRC-047	260–260	Other components are independent of component m and its observations.	C.1
SRC-048	260–260	Averaging a policy over the other components preserves the information restrictions and cannot increase the component loss.	C.2
SRC-049	260–260	Optimization separates across components. Apply Theorem 1 to each.	C.3
SRC-050	260–261	Q.E.D. Sum with the positive weights w_m .	C.4
SRC-051	300–300	Define $V_\delta(T) = B'(T) = \sum_m A_m e^{-\gamma_m T}$.	R.1
SRC-052	300–304	Differentiate (29): $C'(T) = [H(T) - c]/T^2$, where $H(T) = B(T) - TV_\delta(T)$.	R.2
SRC-053	304–304	$H'(T) = -TV'_\delta(T) > 0$ for $T > 0$, and H increases from zero to c_{crit} .	R.3
SRC-054	304–304	ASSUME $0 < c < c_{\text{crit}}$. PROVE a unique minimizing positive period and its displayed value.	R.4, R.4.1
SRC-055	304–304	$C(T) \rightarrow \infty$ as $T \downarrow 0$ and $C(T) \rightarrow J_L^{\text{ot}}$ as $T \rightarrow \infty$, proving the unique global periodic minimum.	R.4.2
SRC-056	304–304	At the root, $c - B(T^*) = -T^* V_\delta(T^*)$, giving the optimal-cost formula.	R.4.3
SRC-057	304–304	Q.E.D. The below-threshold periodic claims hold.	R.4.4

Source ID	Lines	Source claim or transition	Placement
SRC-058	304–306	If $c \geq c_{\text{crit}}$, then $c - B(T) \geq 0$ for each finite T , so no periodic schedule improves on no refreshing.	R.5
SRC-059	306–306	Every locally finite independent schedule has average cost at least the minimum of the periodic costs and the no-refresh cost. The stated candidates attain this bound.	R.6
SRC-060	306–306	Define $g = \min\{0, \inf_{T>0}(c - B(T))/T\}$ and condition on a realization of the reception schedule.	R.6.1
SRC-061	306–306	Every complete interval of length T_j contributes at least gT_j relative to local operation when assigned one communication price.	R.6.2
SRC-062	306–306	The last incomplete interval contributes at least $-c_{\text{crit}}$, and the initial interval before the first reception has zero gain.	R.6.3
SRC-063	306–310	Because $g \leq 0$, the expected total cost through time R , conditional on the schedule, is at least $R(J_L^{\text{tot}} + g) - c_{\text{crit}}$.	R.6.4
SRC-064	310–310	Charging transmissions at generation preserves the bound since sent messages are at least as numerous as received messages.	R.6.5
SRC-065	310–310	The bound is uniform over schedules. Take expectation over the schedule, divide by R , and then take the limiting upper average to obtain the independent-randomized-schedule bound.	R.6.6
SRC-066	310–310	Periodic operation at T^* below the threshold, or no communication otherwise, attains the bound.	R.6.7
SRC-067	310–311	Q.E.D. The all-schedules optimum is attained by the stated candidates.	R.6.8, R.7

4 Lamport-style rendering

Each capital-letter prefix identifies a theorem unit. Decimal labels indicate nesting. For example, H.2.7 lies inside the positive-age case H.2. Only a completed parent statement is available after its subproof closes. Each source-support label records what the source supplies; it is not a validity assessment.

4.1 Lemma 1: Team optimality

L.1. For a feasible perturbation w , the objective difference has linear term $2n^{-1} \sum_i \mathbb{E}[w_i((1 + \kappa)u_i - \kappa\bar{u} - X)]$ and a quadratic remainder.

Source: SRC-001. Support recorded in source: EXPLICIT.

L.2. The linear term vanishes for every feasible w if and only if the conditional normal equations (5) hold.

Source: SRC-002. Support recorded in source: EXPLICIT.

L.3. The remainder is $n^{-1} \sum_i \mathbb{E}[w_i^2] + \kappa n^{-1} \sum_i \mathbb{E}[(w_i - \bar{w})^2]$, positive whenever w is nonzero in L^2 .

Source: SRC-003. Support recorded in source: EXPLICIT.

L.4. Q.E.D. Necessity, sufficiency, and uniqueness follow.

Source: SRC-004. Support recorded in source: EXPLICIT.

4.2 Proposition 1: Fresh endpoints

P.1. $X(t) - aY_i(t)/(a+b)$ has zero covariance with every $Y_i(r)$ for $r \leq t$, so it is independent of the local history.

Source: SRC-005. Support recorded in source: EXPLICIT.

P.2. Define $\eta(t) = X(t) - (a/v)\bar{Y}(t)$. It is uncorrelated with every $Y_j(r)$ and independent of the entire sensor process.

Source: SRC-006. Support recorded in source: EXPLICIT.

P.3. The residual variances are $P_1 = ab/(a+b)$ and $P_n = ab/(b+na)$. The corresponding conditional means are $aY_i(t)/(a+b)$ and $(a/v)\bar{Y}(t)$.

Source: SRC-007. Support recorded in source: EXPLICIT.

P.4. For the candidate $u_i = kY_i(t)$, $\mathbb{E}[\bar{u} \mid \mathcal{F}_i(t)] = kvY_i(t)/(a+b)$.

Source: SRC-008. Support recorded in source: EXPLICIT.

P.5. Equation (5) gives $k[(1+\kappa)(a+b) - \kappa v] = a$. The bracket equals d , so $k = a/d$.

Source: SRC-009. Support recorded in source: EXPLICIT.

P.6. The local loss is $a - 2ak + dk^2 = a - a^2/d$.

Source: SRC-010. Support recorded in source: EXPLICIT.

P.7. With pooled information all agents use the common conditional mean, with zero disagreement and loss P_n .

Source: SRC-011. Support recorded in source: EXPLICIT.

P.8. The displayed feasible policies are optimal by Lemma 1.

Source: SRC-012. Support recorded in source: EXPLICIT.

P.9. $d - v = b(1+\kappa)(1 - 1/n) > 0$ and subtraction yields the formula for Δ .

Source: SRC-013. Support recorded in source: EXPLICIT.

P.10. Q.E.D. All endpoint claims follow.

Source: SRC-013. Support recorded in source: IMPLICIT.

4.3 Corollary 1: The delay crossover

D.1. The Markov property and Proposition 1 give $\mathbb{E}[X(t) \mid \mathcal{F}(s)] = e^{-\lambda\tau}(a/v)\bar{Y}(s)$.

Source: SRC-014. Support recorded in source: EXPLICIT.

D.2. Taking this common mean minimizes tracking loss and yields zero disagreement, hence minimizes the team loss.

Source: SRC-015. Support recorded in source: EXPLICIT.

D.3. The error variance is the displayed $J_D(\tau)$.

Source: SRC-016. Support recorded in source: EXPLICIT.

D.4. Comparison with J_L yields $J_L < J_D$ if and only if $e^{-2\lambda\tau} < v/d$.

Source: SRC-017. Support recorded in source: EXPLICIT.

D.5. Taking logarithms gives the exact threshold (14), including its equality case.

Source: SRC-018. Support recorded in source: EXPLICIT.

D.6. Q.E.D.

Source: SRC-018. Support recorded in source: IMPLICIT.

4.4 Theorem 1: Hybrid policy and exponential value

H.1. CASE $\tau = 0$. The formula is the pooled policy.

Source: SRC-019. Support recorded in source: EXPLICIT.

H.2. CASE $\tau > 0$. Prove the remaining claims under this assumption.

Source: SRC-020. Support recorded in source: EXPLICIT.

H.2.1. Enlarge each information field to $\mathcal{G}_i = \mathcal{F}(s) \vee \mathcal{F}_i(t)$ and define $Z_i = Y_i(t) - \rho Y_i(s)$, $h = a/v$, $k = a/d$.

Source: SRC-021. Support recorded in source: EXPLICIT.

H.2.2. Z is independent of $\mathcal{F}(s)$ and has covariance $(1 - \rho^2)(a\mathbf{1}\mathbf{1}^\top + bI)$.

Source: SRC-022. Support recorded in source: EXPLICIT.

H.2.3. $X(t) = \rho h \bar{Y}(s) + h \bar{Z} + \eta(t)$, where $\eta(t)$ is independent of all sensor data and has variance P_n .

Source: SRC-023. Support recorded in source: EXPLICIT.

H.2.4. For $s < r \leq t$, define $Z_i(r) = Y_i(r) - e^{-\lambda(r-s)}Y_i(s)$. Then $\text{Cov}(\bar{Z}, Z_i(r)) = [v/(a + b)] \text{Cov}(Z_i, Z_i(r))$.

Source: SRC-024. Support recorded in source: EXPLICIT.

H.2.5. Both covariances have the temporal factor $e^{-\lambda(t-r)}(1 - e^{-2\lambda(r-s)})$.

Source: SRC-025. Support recorded in source: EXPLICIT.

H.2.6. $\bar{Z} - vZ_i/(a + b)$ is orthogonal to the entire local innovation path and to $\mathcal{F}(s)$.

Source: SRC-026. Support recorded in source: EXPLICIT.

H.2.7. Gaussianity gives $\mathbb{E}[\bar{Z} \mid \mathcal{G}_i] = vZ_i/(a + b)$.

Source: SRC-027. Support recorded in source: EXPLICIT.

H.2.8. Independence of $\eta(t)$ gives $\mathbb{E}[h\bar{Z} + \eta(t) \mid \mathcal{G}_i] = aZ_i/(a + b)$ and $\mathbb{E}[k\bar{Z} \mid \mathcal{G}_i] = kvZ_i/(a + b)$.

Source: SRC-028. Support recorded in source: EXPLICIT.

H.2.9. Subtract the common prediction. Since $(1 + \kappa)(a + b) - \kappa v = d$, the correction kZ_i satisfies (5) under the full path information. Lemma 1 certifies optimality.

Source: SRC-029. Support recorded in source: EXPLICIT.

H.2.10. The policy (15) is feasible under the hybrid field (4), so it attains the same value as complete delayed sharing.

Source: SRC-030. Support recorded in source: EXPLICIT.

H.2.11. The irreducible variance is P_n and the remaining quadratic cost scales by $1 - \rho^2$. Thus $J_H = P_n + (1 - \rho^2)(J_L - P_n)$.

Source: SRC-031. Support recorded in source: EXPLICIT.

H.2.12. The enlarged-information posterior mean is $\rho h \bar{Y}(s) + aZ_i/(a + b)$. It is hybrid-feasible and has error variance $P_n + (1 - \rho^2)(P_1 - P_n)$.

Source: SRC-032. Support recorded in source: EXPLICIT.

H.2.13. Subtracting the cost from J_L yields $\Delta e^{-2\lambda\tau}$.

Source: SRC-033. Support recorded in source: EXPLICIT.

H.2.14. Q.E.D. The positive-age branch is established.

Source: SRC-033. Support recorded in source: IMPLICIT.

H.3. Q.E.D. The two cases cover every $\tau \geq 0$.

Source: SRC-033. Support recorded in source: IMPLICIT.

4.5 Proposition 2: Unequal sensor quality

U.1. Substitution into (5) gives $C_i k_i - \kappa a \bar{k} = a$, where $\bar{k} = n^{-1} \sum_i k_i$.

Source: SRC-034. Support recorded in source: EXPLICIT.

U.2. Averaging $k_i = (a + \kappa a \bar{k})/C_i$ yields the displayed coefficients.

Source: SRC-035. Support recorded in source: EXPLICIT.

U.3. The denominator is positive because $C_i > (1 + \kappa)a$.

Source: SRC-036. Support recorded in source: EXPLICIT.

U.4. Lemma 1 certifies optimality of the local policy.

Source: SRC-037. Support recorded in source: EXPLICIT.

U.5. The quadratic term equals $a\bar{k}$, so $J_L = a - a\bar{k}$.

Source: SRC-038. Support recorded in source: EXPLICIT.

U.6. Gaussian conditioning gives P and μ_P .

Source: SRC-039. Support recorded in source: EXPLICIT.

U.7. $X(t) - \mu_P(t)$ is independent of the entire sensor process because all temporal covariances share one factor.

Source: SRC-040. Support recorded in source: EXPLICIT.

U.8. Innovations have covariance $(1 - \rho^2)\Sigma$, where $\Sigma = a\mathbf{1}\mathbf{1}^\top + \text{diag}(b_i)$.

Source: SRC-041. Support recorded in source: EXPLICIT.

U.9. For $\mathcal{G}_i = \mathcal{F}(s) \vee \mathcal{F}_i(t)$, the covariance argument in (21) gives $\mathbb{E}[Z_j | \mathcal{G}_i] = \Sigma_{ji}Z_i/\Sigma_{ii}$.

Source: SRC-042. Support recorded in source: EXPLICIT.

U.10. After subtracting $\rho\mu_P(s)$, the target has conditional mean $aZ_i/(a + b_i)$.

Source: SRC-043. Support recorded in source: EXPLICIT.

U.11. The residual normal equations are the fresh-local equations, so k_iZ_i is optimal.

Source: SRC-044. Support recorded in source: EXPLICIT.

U.12. Scaling the remaining quadratic cost by $1 - \rho^2$ gives $J_H = \rho^2J_P + (1 - \rho^2)J_L$.

Source: SRC-045. Support recorded in source: EXPLICIT.

U.13. Q.E.D.

Source: SRC-046. Support recorded in source: EXPLICIT.

4.6 Corollary 2: Additive coordination value

C.1. Other components are independent of component m and its observations.

Source: SRC-047. Support recorded in source: EXPLICIT.

C.2. Averaging a policy over the other components preserves the information restrictions and cannot increase the component loss.

Source: SRC-048. Support recorded in source: EXPLICIT.

C.3. Optimization separates across components. Apply Theorem 1 to each.

Source: SRC-049. Support recorded in source: EXPLICIT.

C.4. Q.E.D. Sum with the positive weights w_m .

Source: SRC-050. Support recorded in source: EXPLICIT.

4.7 Theorem 2: Optimal refresh period

R.1. Define $V_\delta(T) = B'(T) = \sum_m A_m e^{-\gamma_m T}$.

Source: SRC-051. Support recorded in source: EXPLICIT.

R.2. Differentiate (29): $C'(T) = [H(T) - c]/T^2$, where $H(T) = B(T) - TV_\delta(T)$.

Source: SRC-052. Support recorded in source: EXPLICIT.

R.3. $H'(T) = -TV'_\delta(T) > 0$ for $T > 0$, and H increases from zero to c_{crit} .

Source: SRC-053. Support recorded in source: EXPLICIT.

R.4. ASSUME $0 < c < c_{\text{crit}}$. PROVE a unique minimizing positive period and its displayed value.

Source: SRC-054. Support recorded in source: EXPLICIT.

R.4.1. There is a unique root $H(T^*) = c$, and C decreases before it and increases after it.

Source: SRC-054. Support recorded in source: EXPLICIT.

R.4.2. $C(T) \rightarrow \infty$ as $T \downarrow 0$ and $C(T) \rightarrow J_L^{\text{tot}}$ as $T \rightarrow \infty$, proving the unique global periodic minimum.

Source: SRC-055. Support recorded in source: EXPLICIT.

R.4.3. At the root, $c - B(T^*) = -T^*V'_\delta(T^*)$, giving the optimal-cost formula.

Source: SRC-056. Support recorded in source: EXPLICIT.

R.4.4. Q.E.D. The below-threshold periodic claims hold.

Source: SRC-057. Support recorded in source: EXPLICIT.

R.5. If $c \geq c_{\text{crit}}$, then $c - B(T) \geq 0$ for each finite T , so no periodic schedule improves on no refreshing.

Source: SRC-058. Support recorded in source: EXPLICIT.

R.6. Every locally finite independent schedule has average cost at least the minimum of the periodic costs and the no-refresh cost. The stated candidates attain this bound.

Source: SRC-059. Support recorded in source: EXPLICIT.

R.6.1. Define $g = \min\{0, \inf_{T>0}(c - B(T))/T\}$ and condition on a realization of the reception schedule.

Source: SRC-060. Support recorded in source: EXPLICIT.

R.6.2. Every complete interval of length T_j contributes at least gT_j relative to local operation when assigned one communication price.

Source: SRC-061. Support recorded in source: EXPLICIT.

R.6.3. The last incomplete interval contributes at least $-c_{\text{crit}}$, and the initial interval before the first reception has zero gain.

Source: SRC-062. Support recorded in source: EXPLICIT.

R.6.4. Because $g \leq 0$, the expected total cost through time R , conditional on the schedule, is at least $R(J_L^{\text{tot}} + g) - c_{\text{crit}}$.

Source: SRC-063. Support recorded in source: EXPLICIT.

R.6.5. Charging transmissions at generation preserves the bound since sent messages are at least as numerous as received messages.

Source: SRC-064. Support recorded in source: EXPLICIT.

R.6.6. The bound is uniform over schedules. Take expectation over the schedule, divide by R , and then take the limiting upper average to obtain the independent-randomized-schedule bound.

Source: SRC-065. Support recorded in source: EXPLICIT.

R.6.7. Periodic operation at T^* below the threshold, or no communication otherwise, attains the bound.

Source: SRC-066. Support recorded in source: EXPLICIT.

R.6.8. Q.E.D. The all-schedules optimum is attained by the stated candidates.

Source: SRC-067. Support recorded in source: EXPLICIT.

R.7. Q.E.D. Both price regimes and every stated policy class are covered.

Source: SRC-067. Support recorded in source: IMPLICIT.

5 Source-to-step mapping ledger

NORMALIZED means that notation or syntax was standardized without changing the mathematical claim. STRUCTURAL records organization already present in the source, such as its zero/positive-age cases or final conclusion. EXPLICIT and IMPLICIT describe source support. No source assertion has been replaced by a newly invented supporting lemma.

Step/item	Source	Kind	Support	Issue	Notes
L	THM-L	NORMALIZED	EXPLICIT	None	Letter lines 102–110.
P	THM-P	NORMALIZED	EXPLICIT	None	Letter lines 117–133.
D	THM-D	NORMALIZED	EXPLICIT	None	Letter lines 155–167.
H	THM-H	NORMALIZED	EXPLICIT	None	Letter lines 175–191.
U	THM-U	NORMALIZED	EXPLICIT	None	Letter lines 230–244.
C	THM-C	NORMALIZED	EXPLICIT	None	Letter lines 252–259.
R	THM-R	NORMALIZED	EXPLICIT	None	Letter lines 284–299.
L.1	SRC-001	NORMALIZED	EXPLICIT	None	Parent L.
L.2	SRC-002	NORMALIZED	EXPLICIT	None	Parent L.
L.3	SRC-003	NORMALIZED	EXPLICIT	None	Parent L.
L.4	SRC-004	STRUCTURAL	EXPLICIT	None	Parent L.
P.1	SRC-005	NORMALIZED	EXPLICIT	None	Parent P.
P.2	SRC-006	NORMALIZED	EXPLICIT	None	Parent P.
P.3	SRC-007	NORMALIZED	EXPLICIT	None	Parent P.
P.4	SRC-008	NORMALIZED	EXPLICIT	None	Parent P.
P.5	SRC-009	NORMALIZED	EXPLICIT	None	Parent P.
P.6	SRC-010	NORMALIZED	EXPLICIT	None	Parent P.
P.7	SRC-011	NORMALIZED	EXPLICIT	None	Parent P.
P.8	SRC-012	NORMALIZED	EXPLICIT	None	Parent P.
P.9	SRC-013	NORMALIZED	EXPLICIT	None	Parent P.
P.10	SRC-013	STRUCTURAL	IMPLICIT	None	Parent P.
D.1	SRC-014	NORMALIZED	EXPLICIT	None	Parent D.

Step/item	Source	Kind	Support	Issue	Notes
D.2	SRC-015	NORMALIZED	EXPLICIT	None	Parent D.
D.3	SRC-016	NORMALIZED	EXPLICIT	None	Parent D.
D.4	SRC-017	NORMALIZED	EXPLICIT	None	Parent D.
D.5	SRC-018	NORMALIZED	EXPLICIT	None	Parent D.
D.6	SRC-018	STRUCTURAL	IMPLICIT	None	Parent D.
H.1	SRC-019	STRUCTURAL	EXPLICIT	None	Parent H.
H.2	SRC-020	STRUCTURAL	EXPLICIT	None	Parent H.
H.2.1	SRC-021	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.2	SRC-022	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.3	SRC-023	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.4	SRC-024	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.5	SRC-025	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.6	SRC-026	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.7	SRC-027	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.8	SRC-028	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.9	SRC-029	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.10	SRC-030	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.11	SRC-031	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.12	SRC-032	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.13	SRC-033	NORMALIZED	EXPLICIT	None	Parent H.2.
H.2.14	SRC-033	STRUCTURAL	IMPLICIT	None	Parent H.2.
H.3	SRC-033	STRUCTURAL	IMPLICIT	None	Parent H.
U.1	SRC-034	NORMALIZED	EXPLICIT	None	Parent U.
U.2	SRC-035	NORMALIZED	EXPLICIT	None	Parent U.
U.3	SRC-036	NORMALIZED	EXPLICIT	None	Parent U.
U.4	SRC-037	NORMALIZED	EXPLICIT	None	Parent U.
U.5	SRC-038	NORMALIZED	EXPLICIT	None	Parent U.
U.6	SRC-039	NORMALIZED	EXPLICIT	None	Parent U.
U.7	SRC-040	NORMALIZED	EXPLICIT	None	Parent U.
U.8	SRC-041	NORMALIZED	EXPLICIT	None	Parent U.
U.9	SRC-042	NORMALIZED	EXPLICIT	None	Parent U.
U.10	SRC-043	NORMALIZED	EXPLICIT	None	Parent U.
U.11	SRC-044	NORMALIZED	EXPLICIT	None	Parent U.
U.12	SRC-045	NORMALIZED	EXPLICIT	None	Parent U.
U.13	SRC-046	STRUCTURAL	EXPLICIT	None	Parent U.
C.1	SRC-047	NORMALIZED	EXPLICIT	None	Parent C.
C.2	SRC-048	NORMALIZED	EXPLICIT	None	Parent C.
C.3	SRC-049	NORMALIZED	EXPLICIT	None	Parent C.
C.4	SRC-050	STRUCTURAL	EXPLICIT	None	Parent C.
R.1	SRC-051	NORMALIZED	EXPLICIT	None	Parent R.
R.2	SRC-052	NORMALIZED	EXPLICIT	None	Parent R.
R.3	SRC-053	NORMALIZED	EXPLICIT	None	Parent R.

Step/item	Source	Kind	Support	Issue	Notes
R.4	SRC-054	STRUCTURAL	EXPLICIT	None	Parent R.
R.4.1	SRC-054	NORMALIZED	EXPLICIT	None	Parent R.4.
R.4.2	SRC-055	NORMALIZED	EXPLICIT	None	Parent R.4.
R.4.3	SRC-056	NORMALIZED	EXPLICIT	None	Parent R.4.
R.4.4	SRC-057	STRUCTURAL	EXPLICIT	None	Parent R.4.
R.5	SRC-058	NORMALIZED	EXPLICIT	None	Parent R.
R.6	SRC-059	NORMALIZED	EXPLICIT	None	Parent R.
R.6.1	SRC-060	NORMALIZED	EXPLICIT	None	Parent R.6.
R.6.2	SRC-061	NORMALIZED	EXPLICIT	None	Parent R.6.
R.6.3	SRC-062	NORMALIZED	EXPLICIT	None	Parent R.6.
R.6.4	SRC-063	NORMALIZED	EXPLICIT	None	Parent R.6.
R.6.5	SRC-064	NORMALIZED	EXPLICIT	None	Parent R.6.
R.6.6	SRC-065	NORMALIZED	EXPLICIT	None	Parent R.6.
R.6.7	SRC-066	NORMALIZED	EXPLICIT	None	Parent R.6.
R.6.8	SRC-067	STRUCTURAL	EXPLICIT	None	Parent R.6.
R.7	SRC-067	STRUCTURAL	IMPLICIT	None	Parent R.

6 Gap and ambiguity register

There are no open, admitted, unavailable-external, or materially ambiguous support items in the converted formal proofs under the declared mathematical background. Consequently, there are no **GAP**, **ADM**, **EXT**, or **AMB** entries or obligation rows to discharge.

This finding is limited to the specified proof boundary. For example, the model does not authorize an extension to different signal and sensor-error rates within one component, or to communication times chosen from the observations. Such extensions are outside the theorem statements rather than hidden premises accepted by this audit.

7 Frozen audit handoff

The theorem contracts, accepted background, rendered steps, source inventory, and mapping ledger are fixed before the two audit ledgers are evaluated. Mapping metadata identifies provenance only. It is not proof evidence.

The review package includes the exact source, the three plugin instruction files, the conversion record, both audit ledgers, and the independently runnable numerical checks. These identifiers make the reviewed versions unambiguous:

- Plugin repository commit:
c5466f44d6ef0fff983ad67a46bc54e39858826f.
- Letter TeX SHA-256:
ca4063836ef506a56fbee54ab850a67499d70fc5a1de8f5c4f8077a1794f8db5.
- Letter PDF SHA-256:
211b1eae40836b0136d35ea59f6678e5b88df1efa8d01bd100a2ee1f229bf0ef.
- Conversion record SHA-256:
a599596db88f9006726337403f992a6d8c033239f7c44a940aa86637d8daf165.

The plugin instructions are available at the pinned repository revision. The project describes these outputs as model-assisted review artifacts, not machine-checked proofs.

Part II

Forward audit

8 Verdict

PASS. The source-supplied proof routes establish the stated results. Every labeled step is supported within its active scope. No critical, major, or minor issue remains unresolved.

9 Theorem contract

The audit uses exactly the seven contracts in Part I. In particular, competitors are all feasible L^2 policies, the hybrid includes the retained local sample, the unequal-variance extension retains the common temporal rate, and the refresh theorem is restricted to schedules independent of the sensor processes.

No proof is treated as establishing the stronger claim that the optimal refresh schedule is unique. The theorem identifies a unique *optimal periodic interval*. Other schedules can have the same long-run cost, for example after changing finitely many refresh times.

10 Proof-structure summary

All generated step labels are unambiguous. The hierarchy has at most two nested levels within a theorem unit. No dependency points to a later step. The positive-age assumption in the hybrid proof is local to its case, and the below-threshold assumption in the periodic optimization subproof is discharged before the general-schedule comparison.

Proof	Steps	Max. level	Main pattern
L	4	1	Quadratic variation and strict convexity.
P	10	1	Gaussian projection, feasible policy, and matching optimality equations.
D	6	1	Common conditional mean and exact scalar comparison.
H	17	2	Zero/positive age cases, full-history Gaussian projection, direct normal equations, and matching feasible policy.
U	13	1	Unequal-coefficient normal equations and covariance-factor extension.
C	4	1	Independence, joint averaging, and an attained additive bound.
R	19	2	Scalar minimization, finite-horizon interval lower bound, and attainment.

The hierarchy contains no unjustified PICK witness. In the refresh proof, continuity, endpoint limits, and strict monotonicity establish the existence and uniqueness of T^* before that root is used. The zero-age hybrid case is covered explicitly, including its vanishing innovation and pooled posterior.

11 Findings and checked derivations

11.1 Findings

No proof defect was identified. The central full-history projection is checked in K-H1–K-H3. Proposition 2 states the generalized covariance projection and invokes Lemma 1. Its coefficient denominators and full-history projections are checked in K-U. Theorem 2 explicitly conditions the expected cost on the schedule and then averages before taking the long-run limit, as checked in K-R.

The remainder of this section supplies the checked algebra, covariance identities, and side conditions behind the ledger. Each calculation follows the operation invoked by the source. None substitutes a different proof route.

11.2 K-L: Normal equations and nonlinear competitors

Let $r_i = (1 + \kappa)u_i - \kappa\bar{u} - X$. All r_i belong to L^2 because X and the actions do. Expanding (A1) gives

$$\begin{aligned} J(u + w) - J(u) &= \frac{2}{n} \sum_i \mathbb{E}[w_i r_i] + Q(w), \\ Q(w) &= \frac{1}{n} \sum_i \mathbb{E}[w_i^2] + \frac{\kappa}{n} \sum_i \mathbb{E}[(w_i - \bar{w})^2]. \end{aligned} \quad (\text{A2})$$

For each feasible w_i , conditional expectation gives

$$\mathbb{E}[w_i r_i] = \mathbb{E}[w_i \mathbb{E}[r_i \mid \mathcal{G}_i]].$$

If the linear term vanishes for all feasible perturbations, choose one nonzero component at a time and set $w_i = \mathbb{E}[r_i \mid \mathcal{G}_i]$. The L^2 contraction property ensures this is admissible. It follows that $\mathbb{E}[r_i \mid \mathcal{G}_i] = 0$ almost surely. Since u_i is $\mathcal{G}_i$ -measurable, these are precisely the source normal equations. Conversely, those equations make every linear term zero.

If the linear functional is nonzero in some feasible direction w , the variation ϵw has cost change $\epsilon L + \epsilon^2 Q(w)$ with $L \neq 0$. A sufficiently small ϵ of the opposite sign improves the objective. This proves necessity. If all linear terms vanish, (A2) is nonnegative, proving sufficiency. Its first sum is strictly positive for a nonzero L^2 perturbation, proving uniqueness whenever an optimum exists. The later propositions explicitly construct the needed optima. The proof never restricts competitors to linear policies.

11.3 K-P: Endpoint projections, costs, and the global residual

Write $q_{tr} = e^{-\lambda|t-r|}$. Independence and the common rate give

$$\begin{aligned} \text{Cov}(X(t), Y_i(r)) &= a q_{tr}, \\ \text{Cov}(Y_i(t), Y_i(r)) &= (a + b) q_{tr}, \\ \text{Cov}(\bar{Y}(t), Y_j(r)) &= v q_{tr}. \end{aligned}$$

Thus

$$\text{Cov}\left(X(t) - \frac{a}{a+b} Y_i(t), Y_i(r)\right) = 0, \quad \text{Cov}\left(X(t) - \frac{a}{v} \bar{Y}(t), Y_j(r)\right) = 0.$$

The second equality holds for every j and every r , including future r . Joint Gaussianity therefore gives independence from the entire indicated observation family. Both predictors are measurable in their

assigned history. Their residual variances are

$$a - \frac{a^2}{a+b} = P_1, \quad a - \frac{a^2}{v} = P_n.$$

The same all-history covariance calculation gives $\mathbb{E}[\bar{Y}(t) | \mathcal{F}_i(t)] = vY_i(t)/(a+b)$. Applying Lemma 1 to $kY_i(t)$ yields

$$k[(1+\kappa)(a+b) - \kappa v] = a, \quad (1+\kappa)(a+b) - \kappa v = d > 0.$$

For this candidate,

$$\mathbb{E}[(kY_i - X)^2] = a - 2ak + k^2(a+b), \quad \frac{1}{n} \sum_i \mathbb{E}[(Y_i - \bar{Y})^2] = b(1 - 1/n).$$

The team objective is therefore $a - 2ak + dk^2$, minimized at $k = a/d$. For pooled information, the common conditional mean minimizes every tracking term and simultaneously makes disagreement zero. Subtraction gives

$$\Delta = \frac{a^2(d-v)}{vd}, \quad d - v = b(1+\kappa)(1 - 1/n) > 0.$$

The residual result **E-eta** used later is now checked under the global model, with no case-specific assumption.

11.4 K-D: The delay threshold and its exact comparison class

The OU signal decomposition is $X(t) = \rho X(s) + \xi$, where $\rho = e^{-\lambda\tau}$ and ξ is independent of the signal and disturbance histories through s . Therefore

$$\mathbb{E}[X(t) | \mathcal{F}(s)] = \rho \mathbb{E}[X(s) | \mathcal{F}(s)] = \rho(a/v)\bar{Y}(s).$$

Its explained variance is $\rho^2 a^2/v$, giving the stated J_D . Since $a, v, d > 0$,

$$J_L < J_D \iff \frac{1}{d} > \frac{\rho^2}{v} \iff e^{-2\lambda\tau} < \frac{v}{d} \iff \tau > \frac{1}{2\lambda} \log \frac{d}{v}.$$

Every equivalence preserves its domain. The last step reverses the inequality when dividing by -2λ . Since $d > v$, the crossover age is strictly positive, and equality holds at that value.

These information sets are not nested: delayed-pool-only operation has other sensors' old data, while local operation has its own current data. The comparison does not say that adding an old pool to a fresh local history makes decisions worse. That enlarged hybrid can ignore the pool and in fact achieves a strictly lower loss at finite age.

11.5 K-H1: The full-history covariance identity

Set $\Sigma = a\mathbf{1}\mathbf{1}^\top + bI$ and $q = 1 - \rho^2$. For an old observation at $r \leq s$,

$$\text{Cov}(Z_i, Y_j(r)) = \Sigma_{ij} \left[e^{-\lambda(t-r)} - e^{-\lambda(t-s)} e^{-\lambda(s-r)} \right] = 0.$$

Also $\text{Cov}(Z) = q\Sigma$. Consequently, the endpoint vector is independent of $\mathcal{F}(s)$. The target identity is

$$X(t) = \rho h \bar{Y}(s) + h \bar{Z} + \eta(t), \quad h = a/v,$$

where $\eta(t)$ is independent of the sensor data and has variance P_n .

For $s < r \leq t$, define the private innovation path $Z_i(r) = Y_i(r) - e^{-\lambda(r-s)}Y_i(s)$. Direct expansion from the four raw cross-covariances gives

$$\text{Cov}(Z_j, Z_i(r)) = \Sigma_{ji}f(r), \quad f(r) = e^{-\lambda(t-r)}(1 - e^{-2\lambda(r-s)}).$$

The own-sensor coefficient is $\Sigma_{ii} = a + b$, and every other sensor has $\Sigma_{ji} = a$. Averaging over j gives

$$\text{Cov}(\bar{Z}, Z_i(r)) = vf(r), \quad \text{Cov}(Z_i, Z_i(r)) = (a + b)f(r).$$

This proves source (21) for every intermediate time and for $r = t$. In particular, the factor for the pooled average is $v/(a + b)$. The factor $a/(a + b)$ applies only to a different sensor $j \neq i$. The source uses the pooled factor correctly. The identity does not divide by $f(r)$, and the model gives $a + b > 0$.

11.6 K-H2: Projection on the entire information field

Let $W_i = \bar{Z} - vZ_i/(a + b)$. The innovation identity in K-H1 shows that W_i is orthogonal to every $Z_i(r)$ for $s < r \leq t$. Both $\bar{Z}$ and Z_i are orthogonal to every observation in $\mathcal{F}(s)$, so the same holds for W_i .

Since $Y_i(s)$ belongs to $\mathcal{F}(s)$, the equality

$$Y_i(r) = Z_i(r) + e^{-\lambda(r-s)}Y_i(s)$$

implies

$$\mathcal{G}_i = \mathcal{F}(s) \vee \mathcal{F}_i(t) = \mathcal{F}(s) \vee \sigma\{Z_i(r) : s < r \leq t\}.$$

The residual W_i and every finite family of these observation generators are jointly Gaussian. Their zero cross-covariances give independence from each finite family. Extending from cylinder events gives independence from $\mathcal{G}_i$. Also $\mathbb{E}[W_i] = 0$ and Z_i is $\mathcal{G}_i$ -measurable. Therefore

$$\mathbb{E}[\bar{Z} \mid \mathcal{G}_i] = \frac{v}{a + b}Z_i. \tag{A3}$$

With $T = h\bar{Z} + \eta(t)$ and $h = a/v$, the independent centered residual $\eta(t)$ contributes zero to this conditional mean. Hence

$$\mathbb{E}[T \mid \mathcal{G}_i] = \frac{a}{a + b}Z_i, \quad \mathbb{E}[k\bar{Z} \mid \mathcal{G}_i] = \frac{kv}{a + b}Z_i.$$

These are conditional means for the full observation path. The argument has checked every generator of that history.

11.7 K-H3: Direct optimality under full local histories

Set $m = \rho h \bar{Y}(s)$ and propose $u_i = m + kZ_i$, where $k = a/d$. The common term m is measurable in every $\mathcal{G}_i$, and the average action is $m + k\bar{Z}$. Thus

$$\begin{aligned} (1 + \kappa)u_i - \kappa\mathbb{E}[\bar{u} \mid \mathcal{G}_i] &= m + k \left[(1 + \kappa) - \frac{\kappa v}{a + b} \right] Z_i \\ &= m + \frac{a}{a + b}Z_i = \mathbb{E}[X(t) \mid \mathcal{G}_i], \end{aligned}$$

where $(1 + \kappa)(a + b) - \kappa v = d > 0$. The candidate is a finite linear combination of square-integrable available observations. Lemma 1 therefore establishes its unique optimality among all admissible policies under the enlarged information, including nonlinear functions of the full histories.

Let J_E denote that enlarged-information optimum. Since $\mathcal{H}_i \subseteq \mathcal{G}_i$, information enlargement gives $J_E \leq J_H$. The candidate uses only $Y_i(t)$, $Y_i(s)$, and $\bar{Y}(s)$, all available under $\mathcal{H}_i$. Hence $J_H \leq J_E$, and equality is attained. No intermediate path observation needs to be discarded from the optimization class to obtain this conclusion.

The same projection also checks the posterior claim. The conditional mean $m + aZ_i/(a+b)$ is hybrid-measurable. By the tower property it is the conditional mean under both the enlarged and hybrid fields. The full-path information consequently has the same posterior error as the stated hybrid implementation.

For the variance calculation in K-H4, the residual target has covariances

$$\text{Cov}(T, Z_i) = aq, \quad \text{Var}(Z_i) = (a+b)q, \quad \text{Cov}(\bar{Z}, Z_i) = vq.$$

11.8 K-H4: Team loss and posterior variance

At a fresh endpoint, $X = h\bar{Y} + \eta$, with η independent of Y . Expanding the loss of the local policy separates it as

$$J_L = P_n + \mathbb{E} \left[\frac{1}{n} \sum_i (kY_i - h\bar{Y})^2 + \frac{\kappa}{n} \sum_i (kY_i - k\bar{Y})^2 \right].$$

Replacing Y by Z multiplies the second, quadratic term by q . The residual variance P_n does not change. This gives

$$J_H = P_n + q(J_L - P_n) = \rho^2 J_P + (1 - \rho^2) J_L.$$

For the posterior, K-H2–K-H3 have already identified the full-history conditional mean. Its error is computed from the endpoint covariances. Since

$$\text{Var}(T) = q(a - P_n) + P_n, \quad \frac{\text{Cov}(T, Z_i)^2}{\text{Var}(Z_i)} = q \frac{a^2}{a+b},$$

the error variance is

$$P_n + q \left(a - P_n - \frac{a^2}{a+b} \right) = P_n + q(P_1 - P_n).$$

The posterior mean is hybrid-feasible, so the same variance holds for the smaller hybrid field. At $\tau = 0$, the source's separate case gives the pooled endpoint directly. At infinite age the expressions converge to the local endpoint; infinite age is used only as a limit, not in a division by q .

11.9 K-U: Unequal sensor variances

With $b_i > 0$, $\text{Cov}(Y_i, Y_j) = a + b_i \mathbf{1}_{i=j}$. For $u_i = k_i Y_i$,

$$\mathbb{E}[\bar{u} \mid \mathcal{F}_i(t)] = \frac{a\bar{k} + b_i k_i/n}{a + b_i} Y_i(t).$$

This is a full-history regression because all cross-time covariances still have the same scalar factor. Substitution in Lemma 1 gives

$$[(1 + \kappa)a + b_i(1 + \kappa - \kappa/n)] k_i - \kappa a \bar{k} = a.$$

The bracket is C_i . Averaging $k_i = (a + \kappa a \bar{k})/C_i$ gives

$$\bar{k} = a\theta + \kappa a \theta \bar{k}, \quad \bar{k} = \frac{a\theta}{1 - \kappa a \theta}.$$

The denominator is positive. If $\kappa > 0$, $C_i > (1 + \kappa)a$ implies $\kappa a \theta < \kappa / (1 + \kappa) < 1$. At $\kappa = 0$, the denominator is one. Lemma 1 certifies optimality of these feasible coefficients. Multiplying the normal equations by k_i and averaging shows that the quadratic objective term equals $a\bar{k}$, hence $J_L = a - a\bar{k}$.

Write $S = \sum_i b_i^{-1}$ and $P = (a^{-1} + S)^{-1}$. For $\mu_P = P \sum_i Y_i / b_i$,

$$\text{Cov}(\mu_P, Y_j) = P(aS + 1) = a, \quad \text{Cov}(X - \mu_P, Y_j) = 0.$$

The same factorization holds at every time offset. Thus $X(t) - \mu_P(t)$ is independent of the entire sensor process and its variance is P .

For the hybrid proof, let $\Sigma = a\mathbf{1}\mathbf{1}^\top + \text{diag}(b_i)$. The same direct innovation expansion gives $\text{Cov}(Z_j, Z_i(r)) = \Sigma_{ji}f(r)$. Thus $Z_j - \Sigma_{ji}Z_i/\Sigma_{ii}$ is orthogonal to the entire local innovation path and to every old observation. As in K-H2, Gaussianity gives

$$\mathbb{E}[Z_j \mid \mathcal{G}_i] = \frac{\Sigma_{ji}}{\Sigma_{ii}} Z_i, \quad \Sigma_{ii} = a + b_i > 0.$$

The residual target is $T = \sum_j (P/b_j) Z_j + \eta(t)$, with $\eta(t) = X(t) - \mu_P(t)$ independent of the sensor data. Consequently,

$$\mathbb{E}[T \mid \mathcal{G}_i] = \frac{P(aS + 1)}{a + b_i} Z_i = \frac{a}{a + b_i} Z_i, \quad \mathbb{E}\left[\frac{1}{n} \sum_j k_j Z_j \mid \mathcal{G}_i\right] = \frac{a\bar{k} + b_i k_i / n}{a + b_i} Z_i.$$

Substitution gives exactly $C_i k_i - \kappa a \bar{k} = a$. Lemma 1 therefore certifies the residual action $k_i Z_i$ under the full history. The policy uses the common scalar $\mu_P(s)$ and the agent's own two readings, so the smaller weighted-summary hybrid attains this optimum. The independent residual contributes variance P , while the remaining quadratic cost scales by $q = 1 - \rho^2$. This yields $J_H = P + q(J_L - P)$. The zero-age case gives $Z_i = 0$ and the pooled policy directly.

11.10 K-C: Independent components and additive value

For a fixed component m , let O_m be all component- m observations and let B contain the observations from all other components. The complete component processes are independent, so B is independent of (O_m, X_m) . A feasible component action has the form $u_{im} = f_{im}(O_{im}, B_i)$, where each argument is restricted to the data available to agent i . Averaging the action vector over the joint law of B gives $\tilde{u}_{im} = \int f_{im}(O_{im}, b_i) dP_{B_i}(b_i)$. This remains measurable in O_{im} and square-integrable by conditional expectation. The target is fixed during averaging, and the team loss is convex in its action vector. Conditional Jensen therefore cannot increase the component loss. The extra observations can be correlated across agents; the averaging uses their joint law.

Every feasible joint policy therefore has weighted loss at least the sum of the independent component minima. Choosing all componentwise optimal hybrid actions attains that lower bound simultaneously. This establishes equality of the optimum with the sum, rather than just an inequality. All sums are finite. Different λ_m across components are allowed, but each component still requires its own signal and sensor disturbances to share one rate. Each component uses its own current reading, retained reading, and shared scalar.

11.11 K-R: Refresh thresholds and optimization over all independent schedules

After reception, a component's gain at elapsed time r is $A_m e^{-\gamma_m r}$. Integrating over a reception interval of length T gives $B(T)$. One price per interval therefore produces the periodic average (29).

The derivative calculation is

$$C'(T) = \frac{-TB'(T) - c + B(T)}{T^2} = \frac{H(T) - c}{T^2}.$$

Since $H'(T) = T \sum_m \gamma_m A_m e^{-\gamma_m T} > 0$, $H(0) = 0$, and $H(\infty) = c_{\text{crit}}$, every price strictly between these limits has a unique positive root. The derivative changes from negative to positive there. As $T \downarrow 0$, $B(T) = T \sum_m A_m + O(T^2)$ and c/T diverges. As $T \rightarrow \infty$, $B(T)$ remains bounded. These facts establish the global periodic minimum and its displayed value. For $c \geq c_{\text{crit}}$, $B(T) < c_{\text{crit}} \leq c$ for every finite T , including $c = c_{\text{crit}}$.

To check the stronger all-schedules claim, condition on a deterministic realization of an independent schedule. Let $r_1 < \dots < r_N \leq R$ be its receptions in the observation horizon, and let

$$g = \min\{0, \inf_{T>0} (c - B(T))/T\}.$$

For every complete interval $T_j = r_{j+1} - r_j$,

$$c - B(T_j) \geq gT_j.$$

The scalar g is finite because $B(T) \leq T \sum_m A_m$, so $(c - B(T))/T \geq -\sum_m A_m$. The final, possibly incomplete interval has total gain at most c_{crit} . The initial interval has no pooled gain. Dropping any unassigned nonnegative communication charges is conservative. Since $\sum_j T_j \leq R$ and $g \leq 0$, the conditional expected cost satisfies

$$\mathbb{E}[\text{accumulated cost through } R \mid \text{schedule}] \geq R(J_L^{\text{tot}} + g) - c_{\text{crit}}. \quad (\text{A4})$$

The no-reception and single-reception cases obey the same inequality. Coincident refreshes, if permitted by a schedule representation, only add positive cost; an interval of length zero contributes $c - B(0) = c \geq 0$.

If prices are charged at generation, the number charged is at least the number received in the horizon. The same lower bound therefore holds. For a random schedule independent of the sensor processes, conditioning leaves their law unchanged, so the pointwise age formula is still valid. Taking expectation in (A4), dividing by R , and then taking a limiting upper average gives the lower bound $J_L^{\text{tot}} + g$. This sequence of operations does not exchange expectation with a limiting upper average. If expected communication counts are infinite, the corresponding total objective is infinite and cannot improve the bound.

A periodic schedule at T^* below the threshold, or no refreshing above it, attains the bound. The finite initial latency changes average cost by a term tending to zero. Therefore the optimization is genuinely over all independent locally finite schedules, not just over periods.

11.12 Derived claims outside the formal theorem environments

The following statements in the letter are checked against the same results and parameter domains. They do not introduce additional proof routes for the formal theorems.

1. **Local disagreement gap, source (12).** Subtracting $P_1 = a - a^2/(a+b)$ from $J_L = a - a^2/d$ gives

$$J_L - P_1 = \frac{\kappa a^2 b(1 - 1/n)}{(a+b)d}.$$

It vanishes at $\kappa = 0$ and is positive at $\kappa > 0$.

2. **Delay-tolerance comparisons after source (14).** For fixed a, b, n, λ , $\partial d / \partial \kappa = b(1 - 1/n) > 0$ and v is fixed, so τ_{cross} increases with κ . For fixed a, b, n, κ , it is proportional to $1/\lambda$ and decreases as the rate increases.

3. **Age budget and half-life, source (22).** For $\alpha \in (0, 1)$, requiring a gain at least $\alpha\Delta$ is equivalent to $e^{-2\lambda\tau} \geq \alpha$. Its exact solution is $\tau \leq \log(1/\alpha)/(2\lambda)$. The value half-life follows by taking $\alpha = 1/2$. The signal's normalized correlation is $e^{-\lambda\tau}$, with correlation half-life $\log 2/\lambda$, exactly twice the value half-life. The coherence time $1/\lambda$ is the lag at correlation e^{-1} .
4. **Discrete-time pooling after source (22).** For independent stationary Gaussian AR(1) processes sharing coefficient $q \in (0, 1)$, the covariance at integer lag k is the corresponding variance times $q^{|k|}$. With $s < t$ integers and $Z_i(r) = Y_i(r) - q^{r-s}Y_i(s)$,

$$\text{Cov}(Z_j(t), Z_i(r)) = \Sigma_{ji}q^{t-r}(1 - q^{2(r-s)}), \quad s < r \leq t.$$

Thus the old-history orthogonality, full-path projection, and normal-equation checks in K-P and K-H carry through with $\rho = q^{t-s}$. At age $k = t - s$, the cost is $q^{2k}J_P + (1 - q^{2k})J_L$, so the gain is Δq^{2k} . Age $k = 0$ gives the pooled endpoint directly. The same covariance factorization checks the unequal-variance formulas. The source's Gaussian, stationarity, independence, and common-coefficient assumptions all remain necessary to this stated proof route.

The letter only transfers the endpoint and pooling laws. For example, over a discrete reception interval of N slots, the accumulated value is the sum $\sum_m A_m(1 - q_m^{2N})/(1 - q_m^2)$ under ages counted from the reception slot. The continuous-time integral and the continuous positive root in Theorem 2 are not asserted verbatim for integer refresh periods.

5. **Multirate effective decay, source (27).** With $V(\tau) = \sum_m w_m \Delta_m e^{-2\lambda_m \tau} > 0$, differentiation gives the displayed weighted mean of $2\lambda_m$. If $\lambda_j < \lambda_k$, the ratio of their remaining contributions equals its age-zero ratio times $e^{2(\lambda_k - \lambda_j)\tau}$, so the slower component's relative share grows. Equal rates give constant relative shares.
6. **Scalar period equation and latency cutoff, source (34)–(35).** For one component, set $x = 2\lambda T^*$ and $\chi = c/c_{\text{crit}}$. The root condition reduces to $1 - (1 + x)e^{-x} = \chi$, whose derivative is $xe^{-x} > 0$ for $x > 0$. As $\chi \uparrow 1$, its unique root diverges. For unit component weight,

$$c \geq \frac{\Delta e^{-2\lambda\delta}}{2\lambda} \iff \delta \geq \max \left\{ 0, \frac{1}{2\lambda} \log \frac{\Delta}{2\lambda c} \right\} \quad (\delta \geq 0).$$

The maximum with zero handles prices already above the zero-latency threshold. With a nonunit weight, that weight must multiply Δ ; the letter explicitly states unit weight for (35).

7. **Minimum-interval constraint after source (35).** Below the price threshold, the objective decreases until T^* and increases afterward. Restricting to $T \geq T_{\min} > 0$ gives the minimizing period $\max\{T_{\min}, T^*\}$. The no-refresh threshold is unchanged, since sufficiently large admissible periods still produce a gain whenever $c < c_{\text{crit}}$.
8. **The measurement-noise qualification in Section II-A.** To check the stated limitation, take a stationary Gaussian signal with variance a and one-step correlation $q > 0$, and independent measurement noise of variance $b > 0$ at each sample. The instantaneous residual satisfies

$$\text{Cov} \left(X_t - \frac{a}{a+b} Y_{i,t}, Y_{i,t-1} \right) = aq - \frac{a}{a+b} aq = \frac{abq}{a+b} > 0.$$

The current reading alone therefore fails to summarize the local history in this example. The usual recursive Gaussian conditional update incorporates older readings, as stated by the source's Kalman-filter qualification. This check concerns the explanatory model limitation, not an extension of a pooling theorem to white noise.

11.13 Boundary and interpretation checks

- The delayed-only optimal actions coincide, so their disagreement cost is zero. For $\tau > \tau_{\text{cross}}$ their tracking loss exceeds the entire optimized local team loss, including the local disagreement penalty. This verifies the introduction’s agreement comparison.
- In the hybrid, the scalar sufficiency claim compares against all raw sensor histories through the pooling time, with each agent’s subsequent local history retained in both information sets. It concerns exact real-valued summaries under the stated Gaussian model.
- $\kappa = 0$ is allowed throughout. The local policy becomes the local posterior mean; pooling still improves estimation for $n \geq 2$.
- $\tau = 0$ gives fresh pooling. The source handles this case explicitly, and the full-path projection identity uses the positive denominator $a + b$.
- At positive finite age, $J_H < J_L$. As age tends to infinity, the hybrid gain tends to zero.
- $c = c_{\text{crit}}$ belongs to the no-refresh regime. Every finite periodic interval has a strictly larger average cost than no refreshing.
- The strict positive-gain claims use $n \geq 2$, $a, b > 0$. They are not asserted for a one-agent or noiseless degenerate model.
- The schedule theorem does not cover event-triggered communication. Conditioning on an observation-dependent schedule can change the sensor law and make silence informative.
- The common within-component rate is used in every full-history residual and projection calculation. Unequal sensor variances preserve it; unequal signal and disturbance rates generally do not.

11.14 Coverage of the displayed equations

Every numbered equation in the letter is accounted for below. Model equations and definitions fix the audit boundary. The remaining entries point to their derivation checks in this report.

Letter tions	equa- Content	Audit location
(1)–(4)	Process model, observations, loss, and information fields	Model and conventions
(5)	Necessary and sufficient team equations	K-L
(6)–(11)	Endpoint definitions, posteriors, policies, costs, and residual	K-P
(12)	Local disagreement gap	Derived claim 1
(13)–(14)	Delayed-only policy, cost, and crossover	K-D
(15)–(21)	Hybrid policy, value, variance, and full-history projection	K-H1–K-H4
(22)	Exact age budget	Derived claim 3
(23)–(25)	Unequal-quality coefficients, posterior, and hybrid policy	K-U
(26)	Additive component value	K-C
(27)	Effective decay rate	Derived claim 5

Letter tions	equa- Content	Audit location
(28)–(33)	Integrated benefit, price threshold, optimal period and cost	K-R
(34)–(35)	Scalar period and latency cutoff	Derived claim 6

11.15 Independent numerical checks

The review adds finite-dimensional checks that do not insert the claimed policy into the optimization solver. For a raw observation vector z_i at each agent and $H = (1 + \kappa)I - (\kappa/n)\mathbf{1}\mathbf{1}^\top$, the solver forms

$$Q_{ij} = H_{ij} \text{Cov}(z_i, z_j), \quad r_i = \text{Cov}(z_i, X), \quad Q\theta = r.$$

The expected loss is then evaluated as $a - 2r^\top \theta/n + \theta^\top Q\theta/n$. The test cases use unequal sensor variances and include complete delayed observations at three old times, together with the agent’s readings at three subsequent times. They check that the extra intermediate and older observations receive zero coefficients beyond the theorem’s sufficient data.

There are 24 parameter sets, each tested with local information and with the augmented history, for 48 systems in total. Four sets have $\kappa = 0$. The covariance matrices are positive definite in these tests. Separate calculations expand the full-path projection cross-covariances from the original process covariances and check the pooled residual. An additional 18 independently solved Gaussian AR(1) systems use $q \in \{0.15, 0.6, 0.95\}$, ages $k \in \{1, 2, 4\}$, and $\kappa \in \{0, 1.7\}$. These include unequal sensor variances and all intermediate local integer-time readings.

The 200 schedule checks cover one to four components, prices below, equal to, and above the threshold, random finite reception patterns, and messages still in transit at the horizon. They check the finite-horizon bound, the optimal-period equation, and the minimum-interval constraint.

Check	Result
Independent Gaussian-team linear systems	48
Largest system dimension	126
Largest expected-loss discrepancy	6.66×10^{-16}
Largest policy-coefficient discrepancy	5.38×10^{-15}
Largest conditional normal-equation residual	1.78×10^{-15}
Largest full-path projection covariance residual	2.22×10^{-16}
Independent discrete-time Gaussian-team systems	18
Largest discrete-time expected-loss discrepancy	5.55×10^{-16}
Independent refresh-schedule checks	200
Smallest finite-horizon lower-bound slack	0.189410
Largest optimal-period equation residual	1.11×10^{-14}

The seed is 260906351. The package includes `check_audit.py` and its full parameter-level output `audit_checks.json`. The script requires Python 3, NumPy, and SciPy. The installed package versions are recorded in `audit_checks.json`. Run `python3 check_audit.py` to reproduce these checks. These checks corroborate the algebra and finite-dimensional consequences. They do not prove an all-history or all-schedules theorem. Those claims are assessed by the mathematical steps above.

The letter’s representative values also agree with direct substitution:

$$J_L = \frac{15}{23} \simeq 0.6522, \quad J_P = \frac{1}{9}, \quad \Delta = \frac{112}{207}, \quad \lambda\tau_{\text{cross}} \simeq 0.4691.$$

At $\lambda\tau = 1$, $J_D \simeq 0.8797$, so the local reduction relative to delayed-only loss is about 25.9%. At $\tau = 0.5$ and $\lambda = 1$, $J_H \simeq 0.4531$. For $\delta = 0.1$, the scalar critical price is about 0.2215, and at half that price the optimal period is about 0.8392 with average cost 0.5695.

For the second component $(a_2, b_2, \lambda_2) = (0.4, 0.8, 0.12)$, with $n = 8$, $\kappa_2 = 1$, and both weights one, direct evaluation gives a joint threshold 1.1807. The second component contributes 34.2% of the value at reception and 81.2% of its lifetime integral. At half the joint threshold, the optimal period is 5.5513 and the total cost is 0.9072. The source package also includes its 83-system parameter records and figure-generation code, so these illustration checks can be reproduced separately from this audit’s augmented-history checks.

12 Forward step ledger

There is one row for every labeled step, including parent claims and internal closing steps. A goal G-H means the full contract for Theorem 1. A goal H.2 means that contract under its positive-age case assumption. The enclosing assumptions and definitions remain active within that goal and are discharged when its parent closes. The calculation IDs refer to the preceding section.

12.1 Lemma 1: Team optimality

Step	Goal	Legal dependencies	Justification checked	Status
L.1	G-L	A-L2, D-loss	K-L: expansion is an identity for square-integrable policies.	OK
L.2	G-L	L.1, B-CE	K-L: conditional expectation converts testing against every local perturbation into a zero conditional residual.	OK
L.3	G-L	A-L2, D-loss	K-L: the first sum is strictly positive and the second nonnegative.	OK
L.4	G-L	L.1, L.2, L.3	A nonzero linear term admits a small improving signed perturbation; a zero linear term leaves a positive remainder.	OK

12.2 Proposition 1: Fresh endpoints

Step	Goal	Legal dependencies	Justification checked	Status
P.1	G-P	A-G, D-Y, B-G	K-P: the common exponential factor cancels for every earlier time.	OK
P.2	G-P	A-G, D-Y, B-G	K-P: $\text{Cov}(\bar{Y}(t), Y_j(r)) = v \exp(-\lambda t-r)$; this works for past and future r .	OK
P.3	G-P	P.1, P.2, B-CE	K-P: subtract the explained variance from a .	OK
P.4	G-P	A-G, D-Y, B-G	K-P: the same regression residual is orthogonal to the full local history.	OK
P.5	G-P	L, P.3, P.4, D-vd	K-P: d is positive under $a, b > 0$ and $\kappa \geq 0$.	OK
P.6	G-P	P.5, D-loss	K-P: $E(Y_i^2) = a + b$ and average expected disagreement of the readings is $b(1-1/n)$.	OK

Step	Goal	Legal dependencies	Justification checked	Status
P.7	G-P	P.3, B-CE, D-loss	Each squared tracking error is separately minimized and the nonnegative disagreement term is zero.	OK
P.8	G-P	L, P.5, P.7	The normal equations are satisfied; linearity was a candidate construction, not a restriction on competitors.	OK
P.9	G-P	A-G, D-vd, P.6, P.7	$n \geq 2$, $b > 0$, and $\kappa \geq 0$ make the strict sign valid.	OK
P.10	G-P	P.3, P.6, P.7, P.8, P.9	Every component of the proposition is included.	OK

12.3 Corollary 1: The delay crossover

Step	Goal	Legal dependencies	Justification checked	Status
D.1	G-D	A-G, P, B-OU, B-CE	K-D: the post-s signal innovation is independent of the old sensor history.	OK
D.2	G-D	D.1, D-loss, B-CE	The lower bound is attained simultaneously for all agents.	OK
D.3	G-D	D.1, D.2, D-vd	K-D: the explained variance is $\exp(-2 \lambda \tau) a^2/v$.	OK
D.4	G-D	D.3, P	Both denominators and a^2 are positive.	OK
D.5	G-D	D.4, A-G	K-D: dividing by -2λ reverses the inequality.	OK
D.6	G-D	D.1, D.2, D.3, D.5	Policy, value, strict comparison, and equality case all close.	OK

12.4 Theorem 1: Hybrid policy and exponential value

Step	Goal	Legal dependencies	Justification checked	Status
H.1	G-H	P	The local correction is zero; the posterior and cost formulas reduce to the pooled endpoints.	OK
H.2	G-H	A-G	All descendants of H.2 have $\tau > 0$ in scope. This assumption is discharged at the root Q.E.D.	OK
H.2.1	H.2	D-info, D-vd	This is an enlargement, so its minimum is a lower bound on the hybrid minimum.	OK
H.2.2	H.2	H.2.1, A-G, B-OU, B-G	K-H1: all old-history cross-covariances cancel.	OK
H.2.3	H.2	H.2.1, E-eta, P	This is the source residual decomposition with $\bar{Y}(t) = \rho \bar{Y}(s) + \bar{Z}$.	OK
H.2.4	H.2	H.2.1, A-G, D-vd	K-H1 expands the two covariances from (1); the pooled factor v includes the own-sensor contribution.	OK
H.2.5	H.2	H.2.4, A-G	K-H1 computes the factor for every intermediate time, including $r=t$.	OK
H.2.6	H.2	H.2.2, H.2.4, H.2.5	K-H2 checks every generator of the enlarged information field. The whole innovation path is independent of the old history.	OK
H.2.7	H.2	H.2.1, H.2.6, B-G, B-CE	K-H2: the regression residual is independent of the generated sigma-field and the proposed predictor is measurable in it.	OK
H.2.8	H.2	H.2.3, H.2.7, B-CE	K-H2: $h \mathbf{v} = \mathbf{a}$, η is centered and independent of all observations, and $\mathbf{k}$ is deterministic.	OK
H.2.9	H.2	H.2.1, H.2.3, H.2.8, D-vd, L	K-H3 substitutes the full-history conditional means directly in the normal equation. All admissible nonlinear history-dependent competitors are covered.	OK

Step	Goal	Legal dependencies	Justification checked	Status
H.2.10	H.2	H.2.1, H.2.9, D-info	The policy only uses the current and retained own readings and the old pooled scalar. Feasibility gives equality of the two optimal values.	OK
H.2.11	H.2	H.2.2, H.2.3, H.2.9, P	K-H4 scales the sensor-dependent quadratic cost and retains the independent residual variance.	OK
H.2.12	H.2	H.2.2, H.2.3, H.2.8, P, B-CE	K-H4 computes the posterior variance and checks that the same conditional mean is measurable in the smaller hybrid field.	OK
H.2.13	H.2	H.2.11, P	The interpolation gives $\rho^2 \Delta$, with $\rho^2 = \exp(-2 \lambda \tau)$.	OK
H.2.14	H.2	H.2.10, H.2.11, H.2.12, H.2.13	Exports the theorem under $\tau > 0$.	OK
H.3	G-H	H.1, H.2, A-G	Uses the completed case statements, with no reference into their private descendants.	OK

12.5 Proposition 2: Unequal sensor quality

Step	Goal	Legal dependencies	Justification checked	Status
U.1	G-U	L, A-U, B-G	K-U: the cross-agent covariance is a and the own variance $a + b_i$; local histories have the same scalar temporal factor.	OK
U.2	G-U	U.1, A-U	K-U solves $\bar{k} = a \theta / (1 - \kappa a \theta)$; denominator positivity follows directly from A-U.	OK
U.3	G-U	A-U	$\kappa a \theta < \kappa / (1 + \kappa) < 1$, with $\kappa = 0$ handled directly.	OK
U.4	G-U	U.1, U.2, U.3, L	The feasible candidate satisfies the conditional equations under the full local histories, so nonlinear competitors are covered.	OK
U.5	G-U	U.1, U.2, U.4, D-loss	K-U: multiply each normal equation by k_i and average.	OK
U.6	G-U	A-U, B-G	K-U verifies the weighted residual is orthogonal to every sensor.	OK
U.7	G-U	U.6, A-U, B-G	The zero same-time cross-covariances remain zero after multiplication by the common temporal factor.	OK
U.8	G-U	A-U, B-OU	The OU innovation calculation applies entry by entry.	OK
U.9	G-U	U.8, A-U, B-G, B-OU	K-U repeats the source covariance argument with Σ ; it treats $j=i$ separately through $\Sigma_{ii} = a + b_i$. All conditional predictors are measurable and denominators positive.	OK
U.10	G-U	U.6, U.7, U.8, U.9, B-CE	K-U uses $P(a \sum_j 1/b_j + 1) = a$ to compute the residual target projection.	OK
U.11	G-U	U.1, U.2, U.4, U.9, U.10, L	K-U substitutes the full-history conditional mean of the weighted action average and verifies the normal equations directly.	OK
U.12	G-U	U.5, U.6, U.7, U.8, U.11, D-loss	K-U keeps the residual variance P fixed, scales only the sensor-dependent quadratic term, and checks feasibility in the weighted-summary hybrid.	OK
U.13	G-U	U.2, U.3, U.4, U.5, U.6, U.11, U.12	All stated coefficients, posteriors, actions, and cost formulas are accounted for.	OK

12.6 Corollary 2: Additive coordination value

Step	Goal	Legal dependencies	Justification checked	Status
C.1	G-C	A-C	This is independence of complete component processes, not just same-time uncorrelatedness.	OK
C.2	G-C	C.1, B-J, D-loss, A-L2	K-C: integrate the joint extraneous component data; each averaged action still depends only on that agent's component-m information.	OK
C.3	G-C	C.2, H, A-C	An additive lower bound is attained by the collection of componentwise optimal policies.	OK
C.4	G-C	C.3, A-C	The sum is finite; no interchange of an infinite series and a limit is involved.	OK

12.7 Theorem 2: Optimal refresh period

Step	Goal	Legal dependencies	Justification checked	Status
R.1	G-R	A-R, D-refresh	All sums are finite and all amplitudes and rates positive.	OK
R.2	G-R	R.1, D-refresh, B-calc	K-R: the quotient rule gives the numerator with this sign.	OK
R.3	G-R	R.1, R.2, A-R, B-calc	K-R: $T \exp(-\gamma T)$ tends to zero for $\gamma > 0$.	OK
R.4	G-R	A-R	This local subproof exports the whole conditional statement.	OK
R.4.1	R.4	R.2, R.3, B-calc	Continuity, strict monotonicity, and the endpoint limits establish existence before the root is used.	OK
R.4.2	R.4	R.4.1, D-refresh, A-R	$c > 0$ gives divergence at zero; B is bounded at infinity.	OK
R.4.3	R.4	R.4.1, D-refresh	$T_{\text{star}} > 0$ licenses division.	OK
R.4.4	R.4	R.4.1, R.4.2, R.4.3	The root and its defining assumptions remain local to R.4; the conditional conclusion is exported.	OK
R.5	G-R	A-R, D-refresh	In fact $B(T) < c_{\text{crit}}$ for every finite T, including the price equality case.	OK
R.6	G-R	A-R	This parent is the all-schedules lower bound and its attainment.	OK
R.6.1	R.6	A-R, D-refresh	g is finite and nonpositive. Independence leaves the conditional sensor-process law unchanged.	OK
R.6.2	R.6	R.6.1, C, D-refresh	K-R: integrate the exact conditional age gain, yielding $c-B(T_{\text{j}})$.	OK
R.6.3	R.6	D-refresh, A-R	$B(L) \leq c_{\text{crit}}$ uniformly in the final interval length L; the source starts without a shared update.	OK
R.6.4	R.6	R.6.1, R.6.2, R.6.3	K-R shows the complete intervals occupy at most R, so multiplication by g reverses the length comparison.	OK
R.6.5	R.6	R.6.4, A-R	The count difference contributes a nonnegative cost.	OK
R.6.6	R.6	R.6.5, A-R, B-CE	K-R verifies the source explicitly takes expectation before limsup. The schedule-independent remainder vanishes after division by R.	OK
R.6.7	R.6	R.4, R.5, R.6.6, D-refresh	Transient latency is finite; periodic time averaging gives (29). Both candidates are admissible.	OK
R.6.8	R.6	R.6.6, R.6.7	Exports the lower bound and matching construction.	OK
R.7	G-R	R.4, R.5, R.6	No event-triggered schedule is included in the information-independent schedule class.	OK

13 Unresolved obligations

None under the stated source and mathematical background. The full-history projections, denominator signs, information constraints, and schedule-expectation order are all supported by the submitted proof routes and checked calculations.

The numerical experiments are not used to discharge a symbolic inference, a continuum-history assertion, or an optimization claim over arbitrary schedules. No external arXiv proof or unavailable cited lemma is needed for closure.

14 Minimal repair plan

No mathematical repair or proof clarification is required by this audit. The source manuscript is unchanged. Its stated common-rate Gaussian model and independent-schedule restriction define the scope of the conclusions.

Part III

Conclusion-first reverse audit

15 Audit boundary

The roots are exactly the seven formal theorem contracts in Part I. The reverse audit uses the same frozen rendering and accepted background, together with the forward step ledger. Each dependency is rechecked for its exact proposition and scope. An earlier theorem is used through its public conclusion, and the explicitly cited residual property is tracked as **E-eta**.

Only routes actually supplied by the paper are represented. There is one primary justification route for each claim. A route is an AND group: every listed obligation is required. The hybrid proof's zero- and positive-age cases are jointly required with their coverage of $\tau \geq 0$; they are not alternative proofs that could rescue a failing case. The conditional branches in the periodic optimization retain their own price assumptions. No speculative or diagnostic alternative proof is used to close a root.

The tables use compact identifiers for exact statements printed in the hierarchy and repeated in the dependency table. Scope conditions belong to their enclosing goals. Thus the claim H.2 is the full theorem under $\tau > 0$, and its closure exports that conditional claim, not an unconditional conclusion derived by division at $\tau = 0$.

16 Obligation graph

Each row gives one claim's primary route and the exact child obligations that must all hold. For a parent with a subproof, the route goes through its internal Q.E.D. The descendants of that Q.E.D. expose the necessary supporting claims without a sideways reference into another closed subproof.

16.1 Lemma 1: Team optimality

Claim	Primary route	AND obligations
L	R-L	L.4
L.1	R-L.1	A-L2, D-loss
L.2	R-L.2	L.1, B-CE
L.3	R-L.3	A-L2, D-loss
L.4	R-L.4	L.1, L.2, L.3

16.2 Proposition 1: Fresh endpoints

Claim	Primary route	AND obligations
P	R-P	P.10
P.1	R-P.1	A-G, D-Y, B-G
P.2	R-P.2	A-G, D-Y, B-G
P.3	R-P.3	P.1, P.2, B-CE

Claim	Primary route	AND obligations
P.4	R-P.4	A-G, D-Y, B-G
P.5	R-P.5	L, P.3, P.4, D-vd
P.6	R-P.6	P.5, D-loss
P.7	R-P.7	P.3, B-CE, D-loss
P.8	R-P.8	L, P.5, P.7
P.9	R-P.9	A-G, D-vd, P.6, P.7
P.10	R-P.10	P.3, P.6, P.7, P.8, P.9

16.3 Corollary 1: The delay crossover

Claim	Primary route	AND obligations
D	R-D	D.6
D.1	R-D.1	A-G, P, B-OU, B-CE
D.2	R-D.2	D.1, D-loss, B-CE
D.3	R-D.3	D.1, D.2, D-vd
D.4	R-D.4	D.3, P
D.5	R-D.5	D.4, A-G
D.6	R-D.6	D.1, D.2, D.3, D.5

16.4 Theorem 1: Hybrid policy and exponential value

Claim	Primary route	AND obligations
H	R-H	H.3
H.1	R-H.1	P
H.2	R-H.2	H.2.14
H.2.1	R-H.2.1	D-info, D-vd
H.2.2	R-H.2.2	H.2.1, A-G, B-OU, B-G
H.2.3	R-H.2.3	H.2.1, E-eta, P
H.2.4	R-H.2.4	H.2.1, A-G, D-vd
H.2.5	R-H.2.5	H.2.4, A-G
H.2.6	R-H.2.6	H.2.2, H.2.4, H.2.5
H.2.7	R-H.2.7	H.2.1, H.2.6, B-G, B-CE
H.2.8	R-H.2.8	H.2.3, H.2.7, B-CE
H.2.9	R-H.2.9	H.2.1, H.2.3, H.2.8, D-vd, L
H.2.10	R-H.2.10	H.2.1, H.2.9, D-info
H.2.11	R-H.2.11	H.2.2, H.2.3, H.2.9, P
H.2.12	R-H.2.12	H.2.2, H.2.3, H.2.8, P, B-CE
H.2.13	R-H.2.13	H.2.11, P
H.2.14	R-H.2.14	H.2.10, H.2.11, H.2.12, H.2.13
H.3	R-H.3	H.1, H.2, A-G

16.5 Proposition 2: Unequal sensor quality

Claim	Primary route	AND obligations
U	R-U	U.13
U.1	R-U.1	L, A-U, B-G
U.2	R-U.2	U.1, A-U
U.3	R-U.3	A-U
U.4	R-U.4	U.1, U.2, U.3, L
U.5	R-U.5	U.1, U.2, U.4, D-loss
U.6	R-U.6	A-U, B-G
U.7	R-U.7	U.6, A-U, B-G
U.8	R-U.8	A-U, B-OU
U.9	R-U.9	U.8, A-U, B-G, B-OU
U.10	R-U.10	U.6, U.7, U.8, U.9, B-CE
U.11	R-U.11	U.1, U.2, U.4, U.9, U.10, L
U.12	R-U.12	U.5, U.6, U.7, U.8, U.11, D-loss
U.13	R-U.13	U.2, U.3, U.4, U.5, U.6, U.11, U.12

16.6 Corollary 2: Additive coordination value

Claim	Primary route	AND obligations
C	R-C	C.4
C.1	R-C.1	A-C
C.2	R-C.2	C.1, B-J, D-loss, A-L2
C.3	R-C.3	C.2, H, A-C
C.4	R-C.4	C.3, A-C

16.7 Theorem 2: Optimal refresh period

Claim	Primary route	AND obligations
R	R-R	R.7
R.1	R-R.1	A-R, D-refresh
R.2	R-R.2	R.1, D-refresh, B-calc
R.3	R-R.3	R.1, R.2, A-R, B-calc
R.4	R-R.4	R.4.4
R.4.1	R-R.4.1	R.2, R.3, B-calc
R.4.2	R-R.4.2	R.4.1, D-refresh, A-R
R.4.3	R-R.4.3	R.4.1, D-refresh
R.4.4	R-R.4.4	R.4.1, R.4.2, R.4.3
R.5	R-R.5	A-R, D-refresh
R.6	R-R.6	R.6.8
R.6.1	R-R.6.1	A-R, D-refresh

Claim	Primary route	AND obligations
R.6.2	R-R.6.2	R.6.1, C, D-refresh
R.6.3	R-R.6.3	D-refresh, A-R
R.6.4	R-R.6.4	R.6.1, R.6.2, R.6.3
R.6.5	R-R.6.5	R.6.4, A-R
R.6.6	R-R.6.6	R.6.5, A-R, B-CE
R.6.7	R-R.6.7	R.4, R.5, R.6.6, D-refresh
R.6.8	R-R.6.8	R.6.6, R.6.7
R.7	R-R.7	R.4, R.5, R.6

17 Critical findings

There is no failing or open conclusion-reachable leaf in this graph. The delicate obligations are discharged as follows:

- The full-history endpoint reductions require orthogonality to every relevant time and every relevant sensor, not just current-time regression. K-P and K-H1 check those quantifiers.
- The full-path conditional mean requires independence of the regression residual from the generated observation sigma-field, together with measurability of the predictor. K-H2 checks both at H.2.6–H.2.8.
- Optimality requires the full-history normal equations and a feasible policy that attains the lower bound. H.2.9 and H.2.10 provide these two steps, and Lemma 1 covers all admissible policies.
- The unequal-quality extension states its generalized covariance projection. K-U checks U.9–U.12 with the new covariance matrix and the same positive temporal factor.
- The refresh theorem requires a lower bound for every allowed schedule and a schedule that attains it. R.6 supplies both, with a uniform finite-horizon remainder and the correct expectation order.

Every forward step used by a reverse edge has the same exact proposition and active parameter scope. All conclusion-reachable leaves are discharged under the declared source boundary. The covariance example for white measurement noise only illustrates the model boundary and is not a counterexample to a stated theorem.

18 Dependency table

The graph has 96 unique conclusion-reachable nodes. Repeated background obligations and earlier theorem roots are shared. Every node has an exact statement, parent uses, route, classification, and status. “Discharged” means supported in this structured mathematical audit, not certified by a proof-assistant kernel.

ID	Exact claim/obligation	Needed by	Route / edge	Classification	Support	Status
A-G	$n \geq 2$, $a, b, \lambda > 0$, $\kappa \geq 0$; mutually independent, stationary, mean-zero Gaussian signal and disturbances with the common exponential covariance (1).	P. 1, P. 2, P. 4, P. 9, D. 1, D. 5, H. 2. 2, H. 2. 4, H. 2. 5, H. 3	Leaf	hypothesis	Boundary / contract.	<i>Discharged.</i>
A-L2	Admissible actions and perturbations are measurable with respect to their assigned information and have finite second moment. $X \in L^2$ under (1).	L. 1, L. 3, C. 2	Leaf	hypothesis	Boundary / contract.	<i>Discharged.</i>
A-U	The A-G assumptions hold with disturbance variances $b_i > 0$ and unchanged common rate. The definitions C_i, θ are those in (23).	U. 1, U. 2, U. 3, U. 6, U. 7, U. 8, U. 9	Leaf	hypothesis	Boundary / contract.	<i>Discharged.</i>
A-C	A finite number of mutually independent component processes with weights $w_m > 0$, componentwise common rates $\lambda_m > 0$, and ages $\tau_m \geq 0$.	C. 1, C. 3, C. 4	Leaf	hypothesis	Boundary / contract.	<i>Discharged.</i>
A-R	Fixed finite latency, exact joint vectors of componentwise pooled means sampled at the same time, price $c > 0$, no queueing, and schedules independent of the sensor processes. Local finiteness means finitely many transmissions in each bounded interval. The proof uses an initially unshared state.	R. 1, R. 3, R. 4. 2, R. 5, R. 6. 1, R. 6. 3, R. 6. 5, R. 6. 6	Leaf	hypothesis	Boundary / contract.	<i>Discharged.</i>
D-loss	$\ell(u, X) = n^{-1} \sum_i (u_i - X)^2 + \kappa n^{-1} \sum_i (u_i - \bar{u})^2$ and $\bar{u} = n^{-1} \sum_i u_i$.	L. 1, L. 3, P. 6, P. 7, D. 2, U. 5, U. 12, C. 2	Leaf	definition	Boundary / contract.	<i>Discharged.</i>
D-Y	$Y_i = X + E_i$ and $\bar{Y} = n^{-1} \sum_i Y_i$.	P. 1, P. 2, P. 4	Leaf	definition	Boundary / contract.	<i>Discharged.</i>
D-vd	$v = a + b/n$ and $d = a + b[1 + \kappa(1 - 1/n)]$.	P. 5, P. 9, D. 3, H. 2. 1, H. 2. 4, H. 2. 9	Leaf	definition	Boundary / contract.	<i>Discharged.</i>
D-info	The local, pooled, delayed-only, and hybrid information fields are exactly those in Section II-B of the letter.	H. 2. 1, H. 2. 10	Leaf	definition	Boundary / contract.	<i>Discharged.</i>
D-refresh	$\gamma_m = 2\lambda_m$, $A_m = w_m \Delta_m e^{-\gamma_m \delta}$, $B(T) = \sum_m A_m (1 - e^{-\gamma_m T}) / \gamma_m$, and $C(T) = J_L^{\text{tot}} + [c - B(T)] / T$.	R. 1, R. 2, R. 4. 2, R. 4. 3, R. 5, R. 6. 1, R. 6. 2, R. 6. 3, R. 6. 7	Leaf	definition	Boundary / contract.	<i>Discharged.</i>

ID	Exact claim/obligation	Needed by	Route / edge	Classification	Support	Status
B-G	Gaussian regression and orthogonality: a jointly Gaussian residual with zero cross-covariance with every observation is independent of the observation sigma-field. For a nonsingular Gaussian observation vector, the conditional mean is its linear regression; the residual covariance is the corresponding Schur complement.	P. 1, P. 2, P. 4, H. 2.2, H. 2.7, U. 1, U. 6, U. 7, U. 9	Leaf	named theorem	Boundary / contract.	<i>Discharged.</i>
B-CE	Conditional expectation is the L^2 orthogonal projection. It minimizes mean-square error, is characterized by testing against measurable L^2 variables, satisfies the tower property, and is an L^2 contraction.	L. 2, P. 3, P. 7, D. 1, D. 2, H. 2.7, H. 2.8, H. 2.12, U. 10, R. 6.6	Leaf	named theorem	Boundary / contract.	<i>Discharged.</i>
B-OU	For the stated stationary common-rate Gaussian process, the innovation over $[s, t]$ is independent of the past and has covariance $(1 - e^{-2\lambda(t-s)})\Sigma$. Its cross-covariances are computed directly from (1).	D. 1, H. 2.2, U. 8, U. 9	Leaf	named theorem	Boundary / contract.	<i>Discharged.</i>
B-J	Conditional Jensen: $\phi(\mathbb{E}[U \mid \mathcal{A}]) \leq \mathbb{E}[\phi(U) \mid \mathcal{A}]$ for an integrable vector and convex function when the expectations exist. The audit checks information feasibility separately.	C. 2	Leaf	named theorem	Boundary / contract.	<i>Discharged.</i>
B-calc	Finite-dimensional real algebra and order, differentiation, integration of exponentials, and the intermediate value theorem for continuous scalar functions, with signs and domains checked at each use.	R. 2, R. 3, R. 4. 1	Leaf	algebraic/logical step	Boundary / contract.	<i>Discharged.</i>
E-eta	The globally quantified residual statement accompanying source equation (11): $\eta(t) = X(t) - (a/v)\bar{Y}(t)$ is independent of the entire sensor process and has variance P_n . Theorem 1 explicitly cites this source result. Its covariance calculation is checked separately in K-P.	H. 2.3	Leaf	proved earlier	Boundary / contract.	<i>Discharged.</i>
L. 1	For a feasible perturbation w , the objective difference has linear term $2n^{-1} \sum_i \mathbb{E}[w_i((1 + \kappa)u_i - \kappa\bar{u} - X)]$ and a quadratic remainder.	L. 2, L. 4	R-L. 1	major claim	Forward L. 1.	<i>Discharged.</i>
L. 2	The linear term vanishes for every feasible w if and only if the conditional normal equations (5) hold.	L. 4	R-L. 2	major claim	Forward L. 2.	<i>Discharged.</i>
L. 3	The remainder is $n^{-1} \sum_i \mathbb{E}[w_i^2] + \kappa n^{-1} \sum_i \mathbb{E}[(w_i - \bar{w})^2]$, positive whenever w is nonzero in L^2 .	L. 4	R-L. 3	major claim	Forward L. 3.	<i>Discharged.</i>
L. 4	Q.E.D. Necessity, sufficiency, and uniqueness follow.	L	R-L. 4	major claim	Forward L. 4.	<i>Discharged.</i>
P. 1	$X(t) - aY_i(t)/(a + b)$ has zero covariance with every $Y_i(r)$ for $r \leq t$, so it is independent of the local history.	P. 3	R-P. 1	major claim	Forward P. 1.	<i>Discharged.</i>
P. 2	Define $\eta(t) = X(t) - (a/v)\bar{Y}(t)$. It is uncorrelated with every $Y_j(r)$ and independent of the entire sensor process.	P. 3	R-P. 2	major claim	Forward P. 2.	<i>Discharged.</i>
P. 3	The residual variances are $P_1 = ab/(a + b)$ and $P_n = ab/(b + na)$. The corresponding conditional means are $aY_i(t)/(a + b)$ and $(a/v)\bar{Y}(t)$.	P. 5, P. 7, P. 10	R-P. 3	major claim	Forward P. 3.	<i>Discharged.</i>

ID	Exact claim/obligation	Needed by	Route / edge	Classification	Support	Status
P.4	For the candidate $u_i = kY_i(t)$, $\mathbb{E}[\bar{u} \mid \mathcal{F}_i(t)] = kvY_i(t)/(a+b)$.	P.5	R-P.4	major claim	Forward P.4.	<i>Discharged.</i>
P.5	Equation (5) gives $k[(1+\kappa)(a+b) - \kappa v] = a$. The bracket equals d , so $k = a/d$.	P.6, P.8	R-P.5	major claim	Forward P.5.	<i>Discharged.</i>
P.6	The local loss is $a - 2ak + dk^2 = a - a^2/d$.	P.9, P.10	R-P.6	major claim	Forward P.6.	<i>Discharged.</i>
P.7	With pooled information all agents use the common conditional mean, with zero disagreement and loss P_n .	P.8, P.9, P.10	R-P.7	major claim	Forward P.7.	<i>Discharged.</i>
P.8	The displayed feasible policies are optimal by Lemma 1.	P.10	R-P.8	major claim	Forward P.8.	<i>Discharged.</i>
P.9	$d - v = b(1+\kappa)(1 - 1/n) > 0$ and subtraction yields the formula for Δ .	P.10	R-P.9	major claim	Forward P.9.	<i>Discharged.</i>
P.10	Q.E.D. All endpoint claims follow.	P	R-P.10	major claim	Forward P.10.	<i>Discharged.</i>
D.1	The Markov property and Proposition 1 give $\mathbb{E}[X(t) \mid \mathcal{F}(s)] = e^{-\lambda\tau}(a/v)\bar{Y}(s)$.	D.2, D.3, D.6	R-D.1	major claim	Forward D.1.	<i>Discharged.</i>
D.2	Taking this common mean minimizes tracking loss and yields zero disagreement, hence minimizes the team loss.	D.3, D.6	R-D.2	major claim	Forward D.2.	<i>Discharged.</i>
D.3	The error variance is the displayed $J_D(\tau)$.	D.4, D.6	R-D.3	major claim	Forward D.3.	<i>Discharged.</i>
D.4	Comparison with J_L yields $J_L < J_D$ if and only if $e^{-2\lambda\tau} < v/d$.	D.5	R-D.4	major claim	Forward D.4.	<i>Discharged.</i>
D.5	Taking logarithms gives the exact threshold (14), including its equality case.	D.6	R-D.5	major claim	Forward D.5.	<i>Discharged.</i>
D.6	Q.E.D.	D	R-D.6	major claim	Forward D.6.	<i>Discharged.</i>
H.1	CASE $\tau = 0$. The formula is the pooled policy.	H.3	R-H.1	major claim	Forward H.1.	<i>Discharged.</i>
H.2	CASE $\tau > 0$. Prove the remaining claims under this assumption.	H.3	R-H.2	major claim	Forward H.2.	<i>Discharged.</i>
H.2.1	Enlarge each information field to $\mathcal{G}_i = \mathcal{F}(s) \vee \mathcal{F}_i(t)$ and define $Z_i = Y_i(t) - \rho Y_i(s)$, $h = a/v$, $k = a/d$.	H.2.2, H.2.3, H.2.4, H.2.7, H.2.9, H.2.10	R-H.2.1	major claim	Forward H.2.1.	<i>Discharged.</i>
H.2.2	Z is independent of $\mathcal{F}(s)$ and has covariance $(1 - \rho^2)(a\mathbf{1}\mathbf{1}^\top + bI)$.	H.2.6, H.2.11, H.2.12	R-H.2.2	major claim	Forward H.2.2.	<i>Discharged.</i>
H.2.3	$X(t) = \rho h\bar{Y}(s) + h\bar{Z} + \eta(t)$, where $\eta(t)$ is independent of all sensor data and has variance P_n .	H.2.8, H.2.9, H.2.11, H.2.12	R-H.2.3	major claim	Forward H.2.3.	<i>Discharged.</i>
H.2.4	For $s < r \leq t$, define $Z_i(r) = Y_i(r) - e^{-\lambda(r-s)}Y_i(s)$. Then $\text{Cov}(\bar{Z}, Z_i(r)) = [v/(a+b)]\text{Cov}(Z_i, Z_i(r))$.	H.2.5, H.2.6	R-H.2.4	major claim	Forward H.2.4.	<i>Discharged.</i>
H.2.5	Both covariances have the temporal factor $e^{-\lambda(t-r)}(1 - e^{-2\lambda(r-s)})$.	H.2.6	R-H.2.5	major claim	Forward H.2.5.	<i>Discharged.</i>
H.2.6	$\bar{Z} - vZ_i/(a+b)$ is orthogonal to the entire local innovation path and to $\mathcal{F}(s)$.	H.2.7	R-H.2.6	major claim	Forward H.2.6.	<i>Discharged.</i>
H.2.7	Gaussianity gives $\mathbb{E}[\bar{Z} \mid \mathcal{G}_i] = vZ_i/(a+b)$.	H.2.8	R-H.2.7	major claim	Forward H.2.7.	<i>Discharged.</i>

ID	Exact claim/obligation	Needed by	Route / edge	Classification	Support	Status
H.2.8	Independence of $\eta(t)$ gives $\mathbb{E}[h\bar{Z} + \eta(t) \mid \mathcal{G}_i] = aZ_i/(a+b)$ and $\mathbb{E}[k\bar{Z} \mid \mathcal{G}_i] = kvZ_i/(a+b)$.	H.2.9, H.2.12	R-H.2.8	major claim	Forward H.2.8.	<i>Discharged.</i>
H.2.9	Subtract the common prediction. Since $(1+\kappa)(a+b) - \kappa v = d$, the correction kZ_i satisfies (5) under the full path information. Lemma 1 certifies optimality.	H.2.10, H.2.11	R-H.2.9	major claim	Forward H.2.9.	<i>Discharged.</i>
H.2.10	The policy (15) is feasible under the hybrid field (4), so it attains the same value as complete delayed sharing.	H.2.14	R-H.2.10	major claim	Forward H.2.10.	<i>Discharged.</i>
H.2.11	The irreducible variance is P_n and the remaining quadratic cost scales by $1 - \rho^2$. Thus $J_H = P_n + (1 - \rho^2)(J_L - P_n)$.	H.2.13, H.2.14	R-H.2.11	major claim	Forward H.2.11.	<i>Discharged.</i>
H.2.12	The enlarged-information posterior mean is $\rho h\bar{Y}(s) + aZ_i/(a+b)$. It is hybrid-feasible and has error variance $P_n + (1 - \rho^2)(P_1 - P_n)$.	H.2.14	R-H.2.12	major claim	Forward H.2.12.	<i>Discharged.</i>
H.2.13	Subtracting the cost from J_L yields $\Delta e^{-2\lambda\tau}$.	H.2.14	R-H.2.13	major claim	Forward H.2.13.	<i>Discharged.</i>
H.2.14	Q.E.D. The positive-age branch is established.	H.2	R-H.2.14	major claim	Forward H.2.14.	<i>Discharged.</i>
H.3	Q.E.D. The two cases cover every $\tau \geq 0$.	H	R-H.3	major claim	Forward H.3.	<i>Discharged.</i>
U.1	Substitution into (5) gives $C_i k_i - \kappa a \bar{k} = a$, where $\bar{k} = n^{-1} \sum_i k_i$.	U.2, U.4, U.5, U.11	R-U.1	major claim	Forward U.1.	<i>Discharged.</i>
U.2	Averaging $k_i = (a + \kappa a \bar{k})/C_i$ yields the displayed coefficients.	U.4, U.5, U.11, U.13	R-U.2	major claim	Forward U.2.	<i>Discharged.</i>
U.3	The denominator is positive because $C_i > (1 + \kappa)a$.	U.4, U.13	R-U.3	major claim	Forward U.3.	<i>Discharged.</i>
U.4	Lemma 1 certifies optimality of the local policy.	U.5, U.11, U.13	R-U.4	major claim	Forward U.4.	<i>Discharged.</i>
U.5	The quadratic term equals $a\bar{k}$, so $J_L = a - a\bar{k}$.	U.12, U.13	R-U.5	major claim	Forward U.5.	<i>Discharged.</i>
U.6	Gaussian conditioning gives P and μ_P .	U.7, U.10, U.12, U.13	R-U.6	major claim	Forward U.6.	<i>Discharged.</i>
U.7	$X(t) - \mu_P(t)$ is independent of the entire sensor process because all temporal covariances share one factor.	U.10, U.12	R-U.7	major claim	Forward U.7.	<i>Discharged.</i>
U.8	Innovations have covariance $(1 - \rho^2)\Sigma$, where $\Sigma = a\mathbf{1}\mathbf{1}^\top + \text{diag}(b_i)$.	U.9, U.10, U.12	R-U.8	major claim	Forward U.8.	<i>Discharged.</i>
U.9	For $\mathcal{G}_i = \mathcal{F}(s) \vee \mathcal{F}_i(t)$, the covariance argument in (21) gives $\mathbb{E}[Z_j \mid \mathcal{G}_i] = \Sigma_{ji}Z_i/\Sigma_{ii}$.	U.10, U.11	R-U.9	major claim	Forward U.9.	<i>Discharged.</i>
U.10	After subtracting $\rho\mu_P(s)$, the target has conditional mean $aZ_i/(a+b_i)$.	U.11	R-U.10	major claim	Forward U.10.	<i>Discharged.</i>
U.11	The residual normal equations are the fresh-local equations, so k_iZ_i is optimal.	U.12, U.13	R-U.11	major claim	Forward U.11.	<i>Discharged.</i>
U.12	Scaling the remaining quadratic cost by $1 - \rho^2$ gives $J_H = \rho^2 J_P + (1 - \rho^2)J_L$.	U.13	R-U.12	major claim	Forward U.12.	<i>Discharged.</i>
U.13	Q.E.D.	U	R-U.13	major claim	Forward U.13.	<i>Discharged.</i>
C.1	Other components are independent of component m and its observations.	C.2	R-C.1	major claim	Forward C.1.	<i>Discharged.</i>

ID	Exact claim/obligation	Needed by	Route / edge	Classification	Support	Status
C.2	Averaging a policy over the other components preserves the information restrictions and cannot increase the component loss.	C.3	R-C.2	major claim	Forward C.2.	<i>Discharged.</i>
C.3	Optimization separates across components. Apply Theorem 1 to each.	C.4	R-C.3	major claim	Forward C.3.	<i>Discharged.</i>
C.4	Q.E.D. Sum with the positive weights w_m .	C	R-C.4	major claim	Forward C.4.	<i>Discharged.</i>
R.1	Define $V_\delta(T) = B'(T) = \sum_m A_m e^{-\gamma_m T}$.	R.2, R.3	R-R.1	major claim	Forward R.1.	<i>Discharged.</i>
R.2	Differentiate (29): $C'(T) = [H(T) - c]/T^2$, where $H(T) = B(T) - TV_\delta(T)$.	R.3, R.4.1	R-R.2	major claim	Forward R.2.	<i>Discharged.</i>
R.3	$H'(T) = -TV'_\delta(T) > 0$ for $T > 0$, and H increases from zero to c_{crit} .	R.4.1	R-R.3	major claim	Forward R.3.	<i>Discharged.</i>
R.4	ASSUME $0 < c < c_{\text{crit}}$. PROVE a unique minimizing positive period and its displayed value.	R.6.7, R.7	R-R.4	major claim	Forward R.4.	<i>Discharged.</i>
R.4.1	There is a unique root $H(T^*) = c$, and C decreases before it and increases after it.	R.4.2, R.4.3, R.4.4	R-R.4.1	major claim	Forward R.4.1.	<i>Discharged.</i>
R.4.2	$C(T) \rightarrow \infty$ as $T \downarrow 0$ and $C(T) \rightarrow J_L^{\text{tot}}$ as $T \rightarrow \infty$, proving the unique global periodic minimum.	R.4.4	R-R.4.2	major claim	Forward R.4.2.	<i>Discharged.</i>
R.4.3	At the root, $c - B(T^*) = -T^*V_\delta(T^*)$, giving the optimal-cost formula.	R.4.4	R-R.4.3	major claim	Forward R.4.3.	<i>Discharged.</i>
R.4.4	Q.E.D. The below-threshold periodic claims hold.	R.4	R-R.4.4	major claim	Forward R.4.4.	<i>Discharged.</i>
R.5	If $c \geq c_{\text{crit}}$, then $c - B(T) \geq 0$ for each finite T , so no periodic schedule improves on no refreshing.	R.6.7, R.7	R-R.5	major claim	Forward R.5.	<i>Discharged.</i>
R.6	Every locally finite independent schedule has average cost at least the minimum of the periodic costs and the no-refresh cost. The stated candidates attain this bound.	R.7	R-R.6	major claim	Forward R.6.	<i>Discharged.</i>
R.6.1	Define $g = \min\{0, \inf_{T>0} (c - B(T))/T\}$ and condition on a realization of the reception schedule.	R.6.2, R.6.4	R-R.6.1	major claim	Forward R.6.1.	<i>Discharged.</i>
R.6.2	Every complete interval of length T_j contributes at least gT_j relative to local operation when assigned one communication price.	R.6.4	R-R.6.2	major claim	Forward R.6.2.	<i>Discharged.</i>
R.6.3	The last incomplete interval contributes at least $-c_{\text{crit}}$, and the initial interval before the first reception has zero gain.	R.6.4	R-R.6.3	major claim	Forward R.6.3.	<i>Discharged.</i>
R.6.4	Because $g \leq 0$, the expected total cost through time R , conditional on the schedule, is at least $R(J_L^{\text{tot}} + g) - c_{\text{crit}}$.	R.6.5	R-R.6.4	major claim	Forward R.6.4.	<i>Discharged.</i>
R.6.5	Charging transmissions at generation preserves the bound since sent messages are at least as numerous as received messages.	R.6.6	R-R.6.5	major claim	Forward R.6.5.	<i>Discharged.</i>

ID	Exact claim/obligation	Needed by	Route / edge	Classification	Support	Status
R.6.6	The bound is uniform over schedules. Take expectation over the schedule, divide by R , and then take the limiting upper average to obtain the independent-randomized-schedule bound.	R.6.7, R.6.8	R-R.6.6	major claim	Forward R.6.6.	<i>Discharged.</i>
R.6.7	Periodic operation at T^* below the threshold, or no communication otherwise, attains the bound.	R.6.8	R-R.6.7	major claim	Forward R.6.7.	<i>Discharged.</i>
R.6.8	Q.E.D. The all-schedules optimum is attained by the stated candidates.	R.6	R-R.6.8	major claim	Forward R.6.8.	<i>Discharged.</i>
R.7	Q.E.D. Both price regimes and every stated policy class are covered.	R	R-R.7	major claim	Forward R.7.	<i>Discharged.</i>
L	Exact full contract in Section 2.4: Lemma 1: Team optimality.	P.5, P.8, H.2.9, U.1, U.4, U.11	R-L	root	Boundary / contract.	<i>Discharged.</i>
P	Exact full contract in Section 2.5: Proposition 1: Fresh endpoints.	D.1, D.4, H.1, H.2.3, H.2.11, H.2.12, H.2.13	R-P	root	Boundary / contract.	<i>Discharged.</i>
D	Exact full contract in Section 2.6: Corollary 1: The delay crossover.	Root	R-D	root	Boundary / contract.	<i>Discharged.</i>
H	Exact full contract in Section 2.7: Theorem 1: Hybrid policy and exponential value.	C.3	R-H	root	Boundary / contract.	<i>Discharged.</i>
U	Exact full contract in Section 2.8: Proposition 2: Unequal sensor quality.	Root	R-U	root	Boundary / contract.	<i>Discharged.</i>
C	Exact full contract in Section 2.9: Corollary 2: Additive coordination value.	R.6.2	R-C	root	Boundary / contract.	<i>Discharged.</i>
R	Exact full contract in Section 2.10: Theorem 2: Optimal refresh period.	Root	R-R	root	Boundary / contract.	<i>Discharged.</i>

19 Verdict

FOLLOWS. Every obligation on the source-supplied route to each root L, P, D, H, U, C, and R is discharged. The full-history projection and direct normal-equation checks close at H.2.7 and H.2.9, hybrid feasibility closes at H.2.10, the unequal-variance calculation closes at U.11 and U.12, and the all-schedules bound and attainment close at R.6.

The result applies to the specified six-page letter and its stated assumptions. It supports the proof routes for all seven formal results, together with the derived claims checked in the forward review. This is a full model-assisted Lamport audit, with separate source mapping, forward checking, and conclusion-first dependency analysis. It is not a machine-checked formalization.